

Notes for Stat 111: Introduction to Statistical Inference

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Contents

1	Introduction	3
1.1	Exploring and describing Y	3
1.2	Quantiles	3
2	Models and Estimation	5
2.1	Statistical models and fundamentals	5
2.2	Likelihood	6
2.3	Estimation principles	7
2.4	Evaluating estimators	9
3	Asymptotics	11
3.1	Consistency of estimators	11
3.2	Asymptotic tools	12
4	Maximum Likelihood Estimation	15
4.1	Properties of MLE	15
4.2	Score, information, efficiency	15
4.3	Regularity conditions	17
5	Confidence Intervals	19
5.1	Interval estimators	19
5.2	Constructing confidence intervals	20
5.3	Asymptotic approximations	23
6	Regression	24
6.1	Regression frameworks	24
6.2	Conditional learning: linear regression	24
6.3	Conditional learning: logistic regression	25
6.4	Statistical models of predictive regression	27
6.5	Ordinary least-squares estimator	29
6.6	Unconditional learning: linear projection	31
7	Exponential Families and Data Compression	32
7.1	Sufficient statistics	32
7.2	Rao-Blackwellization	33
7.3	NEFs and EFs	34
8	Hypothesis testing	36
8.1	Fundamentals of hypothesis testing	36
8.2	Carrying out hypothesis tests	37
8.3	p-values	41
8.4	Likelihood-based tests	42

9	Bayesian statistics	45
9.1	Bayesian thinking	45
9.2	Bayesian point estimators	47
9.3	Credible intervals	49
9.4	Computing Bayesian estimators via simulation	49
9.5	Conjugate priors	51
9.6	Predictive posterior	53
9.7	Hierarchical models	54
9.8	Decision theory	55
10	Sampling and resampling	57
10.1	Design-based inference	57
10.2	Sampling designs	57
10.3	Resampling	62
11	Causal inference	65
11.1	Causal framework	65
11.2	Randomized controlled trial	66
11.3	Population-based models for experiments	67
11.4	Design-based approach for experiments	68
11.5	Observational studies	70

1 Introduction

1.1 Exploring and describing Y

We define several simple descriptive statistics to describe our data Y .

Definition: Common descriptive statistics

Suppose we collect data $Y_1, \dots, Y_n = y_1, \dots, y_n$. Then, we define the following:

$$1) \bar{y} = \frac{1}{n} \sum_{j=1}^n y_j \text{ (sample mean)}$$

$$2) s^2 = \frac{1}{n-1} \sum_{j=1}^n (y_j - \bar{y})^2 \text{ (sample variance)}$$

Now, suppose we collect data pairs $(x_1, y_1) \dots, (x_n, y_n)$. Then we define the following:

$$1) s_{x,y} = \frac{1}{n-1} \sum_{j=1}^n (y_j - \bar{y})(x_j - \bar{x}) \text{ (sample covariance)}$$

$$2) b_y = \frac{s_{x,y}}{s_x^2} = \frac{s_{x,y}}{s_{x,x}} \text{ (sample linear regression)}$$

We can also describe the distribution of Y using the data. One natural method is to look at the empirical CDF.

Definition: Empirical CDF

Recall the CDF of a r.v. Y can be written

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{E}[\mathbb{1}(Y \leq y)] \text{ by fund. bridge}$$

Then, using the data, we can estimate this quantity by simply replacing the expectation with a sample mean of indicators! Doing so, we've obtained the empirical CDF.

$$\hat{F}_Y(y) = \frac{1}{n} \sum_{j=1}^n \mathbb{1}(Y_j \leq y)$$

1.2 Quantiles

We've thought in terms of CDFs, but another useful quantity is the quantile function, which allows us to move back and forth between probability space and data space.

Definition: Quantile function

Suppose F_Y be a CDF for the r.v. Y . Then, the quantile function is defined

$$Q_Y(p) = F_Y^{-1}(p) = \min \{y : F_Y(y) \geq p\}$$

So, the quantile is the inverse CDF, and can also be understood as the smallest value of y that makes the CDF obtain a value of p .

We also have the sample equivalent of the quantile function, the sample quantile.

Definition: Sample quantile

Suppose we have data $y_1, \dots, y_n$, which we can order as $y_{(1)}, \dots, y_{(n)}$. Then the sample quantile is defined

$$\hat{Q}(p) = y_{(\lceil np \rceil)}$$

2 Models and Estimation

2.1 Statistical models and fundamentals

Statistical models help us describe the underlying data-generating process of the data we observe: $\mathbf{y} = (y_1, \dots, y_n)$, where y_j is the datapoint for individual j .

Definition: Statistical model

Suppose we obtain data $\mathbf{y} = (y_1, \dots, y_n)$. A statistical model views $\mathbf{y}$ as a crystallization of the r.v.s $\mathbf{Y} = (Y_1, \dots, Y_n)$. Then, the model specifies the joint CDF of the data: $F_{\mathbf{Y}}$.

We can also categorize models into parametric and nonparametric models, which assume different properties of the data-generating process.

Definition: Parametric/nonparametric models

A parametric statistical model is parameterized by a finite-dimensional parameter θ which is an element of the parameter space Θ . This can be written:

$$F_{\mathbf{Y};\theta}(\mathbf{y}) = F_{\mathbf{Y}}(\mathbf{y};\theta) = F_{\mathbf{Y}}(\mathbf{y}|\theta)$$

In contrast, a nonparametric model does not assume a finite-dimensional parameter space.

Here's a simple example of a Gaussian model.

Example 2.1.10: Gaussian model

We define a Gaussian model $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$. So, our parameter $\theta = (\mu, \sigma)$. Then, we can write down the joint PDF of our data

$$\begin{aligned} f_{\mathbf{Y}}(\mathbf{y};\theta) &= \prod_{j=1}^n f_{Y_j}(y_j;\theta) \text{ by independence} \\ &= \frac{1}{\sigma^2(2\pi)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{j=1}^n (y_j - \mu)^2 \right\} \text{ by Normal PDF} \end{aligned}$$

In inference problems, often we'd like to estimate what the underlying parameter θ of our model is based on the data we collect. Before we continue, let's cover a couple of key definitions in reference to estimation.

Definition: Statistic, estima-nd/-te/-tor

Suppose we have data $\mathbf{y} = (y_1, \dots, y_n)$. Then, we define the following terms:

- Statistic $T(\mathbf{Y}), T(\mathbf{y})$: quantity that can be computed from data only
- Estimand θ : quantity we want learn from data
- Estimator $\hat{\theta} = T(\mathbf{Y})$: procedure/formula to estimate θ from data
- Estimate $\hat{\theta} = T(\mathbf{y})$: the realized version of our estimator computed from data

A typical inference problem has the following workflow:

1. Collect data $\mathbf{y} = (y_1, \dots, y_n)$, viewed as crystallizations of $\mathbf{Y} = (Y_1, \dots, Y_n)$.
2. Write down a model $F_{\mathbf{Y}}(\mathbf{y}; \theta)$ for the data.
3. Estimate θ using an estimator $\hat{\theta}$.
4. Evaluate the estimator $\hat{\theta}$.

2.2 Likelihood

The likelihood function allows us to relate the data to our parameter of interest. However, the key idea in statistics is that the data $\mathbf{y}$ are fixed, while the underlying parameter θ is unknown. So, the likelihood function is a function of θ , with the data held constant.

Definition: Likelihood and log-likelihood function

Let $\mathbf{y}$ be the observed value of $\mathbf{Y}$. Then, for a statistical model with parameter θ , the likelihood function is

$$\mathcal{L}(\theta; \mathbf{Y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta),$$

the joint density of the data under the specified model. Since the $\log(x)$ is monotonically increasing in x , we also commonly work with the log-likelihood function:

$$\log \mathcal{L}(\theta; \mathbf{Y}) = \log f_{\mathbf{Y}}(\mathbf{y}; \theta)$$

An important insight is that the likelihood is invariant under rescaling. So, if $\mathcal{L}(\theta; \mathbf{Y})$ specifies the likelihood of some data and model, then $c \cdot \mathcal{L}(\theta; \mathbf{Y})$ is also a perfectly valid likelihood function. This means we can drop multiplicative constants in $\mathcal{L}(\theta)$, and we can drop additive constants in $\log \mathcal{L}(\theta)$.

Suppose we'd like to describe a $\text{Pois}(\lambda)$ in terms of the mean μ , such that the model becomes $\text{Pois}(1/\mu)$. The likelihood is also resistant to reparameterization.

Theorem 2.2.8: Invariance of likelihood function

Consider a likelihood function $\mathcal{L}(\theta; \mathbf{Y})$ and a transformation $g(\theta) = \psi$, such that g is one-to-one. Then,

$$\mathcal{L}(\theta; \mathbf{Y}) = \mathcal{L}(\psi; \mathbf{Y})$$

So, the likelihood function is invariant under reparameterization of θ . This follows simply from the fact that we can write the joint density $\mathcal{L}(\theta; \mathbf{Y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta)$ in terms of θ or $\psi = g(\theta)$, it's still the same function!

Let's do a quick example to show how exactly this plays out!

Example: Poisson likelihood

Suppose we collect data $\mathbf{Y} = (Y_1, \dots, Y_n)$, where the Y_j are i.i.d. $\text{Pois}(\lambda)$. Let us derive the likelihood and the log-likelihood functions. Beginning with the likelihood, we apply the Poisson density.

$$\begin{aligned}\mathcal{L}(\lambda; \mathbf{Y}) &= f_{\mathbf{Y}}(\mathbf{y}; \lambda) \\ &= \prod_{j=1}^n \frac{e^{-\lambda} \lambda^{y_j}}{y_j!} \text{ by ind., Pois PMF} \\ &\propto e^{-n\lambda} \lambda^{\sum_j y_j},\end{aligned}$$

where we've taken out the multiplicative constant in the denominator. Notice, that by invariance, we could just as easily write this in terms of some parameter $\psi = g(\lambda)$, where g is 1:1.

$$\mathcal{L}(\psi; \mathbf{Y}) = e^{-ng^{-1}(\psi)} g^{-1}(\psi)^{\sum_j y_j}$$

Next, we derive the log-likelihood.

$$\log \mathcal{L}(\lambda; \mathbf{Y}) = -n\lambda + \log(\lambda) \sum_{j=1}^n y_j$$

2.3 Estimation principles

So, how can we actually use the foundation we've built to estimate θ from some dataset $\mathbf{y}$? We will introduce two common approaches:

1. Maximum likelihood estimation
2. Method of moments estimation

Definition: Maximum likelihood estimation

Suppose we have data $\mathbf{y}$ for a model parameterized by θ . Then, the maximum likelihood estimate (MLE) is the value of θ that maximizes the likelihood function.

$$\hat{\theta}_{\text{MLE}} = \underset{\theta \in \Theta}{\operatorname{argmax}} \mathcal{L}(\theta; \mathbf{Y})$$

In other words, we're finding the θ that best fits our data based on the likelihood function, or the joint PDF of the data under θ . Note that since that $\log(x)$ is monotonically increasing in x , the MLE can also be found using the log-likelihood, which is often easier.

$$\underset{\theta \in \Theta}{\operatorname{argmax}} \log \mathcal{L}(\theta; \mathbf{Y}) = \underset{\theta \in \Theta}{\operatorname{argmax}} \mathcal{L}(\theta; \mathbf{Y})$$

Let's do a simple example of ML estimation.

Example: MLE for Poisson data

Suppose we collect data $\mathbf{Y} = (Y_1, \dots, Y_n)$, where the Y_j are i.i.d. $\text{Pois}(\lambda)$. We'd like to estimate λ using maximum likelihood estimation. By the example in 2.3, we obtain the log-likelihood.

$$\log \mathcal{L}(\lambda; \mathbf{Y}) = -n\lambda + \log(\lambda) \sum_{j=1}^n y_j$$

We maximize the log-likelihood by setting the first derivative equal to 0.

$$\frac{d}{d\lambda} \log \mathcal{L}(\lambda; \mathbf{Y}) = -n + \frac{1}{\lambda} \sum_{j=1}^n y_j = 0 \Rightarrow \boxed{\hat{\lambda}_{\text{MLE}} = \bar{Y}}$$

We always check that this is a maximum by showing concavity of the log-likelihood function using the second-derivative test!

$$\frac{d^2}{d^2\lambda} \log \mathcal{L}(\lambda; \mathbf{Y}) = -\frac{1}{\lambda^2} \sum_{j=1}^n y_j < 0$$

We will study the properties of the MLE in great detail in Chapter 4. For now, we will move on to the method of moments.

Definition: Method of moments principle

Under the method of moments (MoM) principle, we write our estimand in terms of moments, then estimate by replacing the theoretical moments with sample moments.

Let's take a look at a few examples of the MoM principle in action

Example: MoM for Normal data

Suppose we obtain the data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Our estimand is $\theta = (\mu, \sigma^2)$. Writing our estimand in terms of theoretical moments:

$$\mu = \mathbb{E}[Y_1], \sigma^2 = \mathbb{E}[Y_1^2] - (\mathbb{E}[Y_1])^2$$

Then, by the MoM principle, our MoM estimators are

$$\hat{\mu}_{\text{MoM}} = \frac{1}{n} \sum_{j=1}^n Y_j, \hat{\sigma}_{\text{MoM}}^2 = \frac{1}{n} \sum_{j=1}^n Y_j^2 - \left(\frac{1}{n} \sum_{j=1}^n Y_j \right)^2$$

Example: ECDF as MoM estimator

Suppose we have data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} F$, where F is some CDF. Our estimand is the CDF $F(y)$. Writing our estimand in terms of theoretical moments:

$$F(y) = \mathbb{P}(Y_1 \leq y) = \mathbb{E}[\mathbb{1}(Y_1 \leq y)]$$

Then, by the MoM principle, we simply replace the theoretical moments with sample moments.

$$\hat{F}(y) = \frac{1}{n} \sum_{j=1}^n \mathbb{1}(Y_j \leq y)$$

Note that we've just derived the ECDF!

2.4 Evaluating estimators

But how exactly do we evaluate if an estimator is “good” for estimating θ ?

Definition: Bias and standard error

Suppose we have an estimator $\hat{\theta}$ for an estimand θ . Then, we define the following quantities:

- 1) $\text{bias}(\hat{\theta}; \theta) = \mathbb{E}[\hat{\theta}] - \theta$
- 2) $\text{SE}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$

An estimator is called unbiased if its bias is 0.

Another common evaluation metric is the MSE, which has a nice interpretation as a combination of the bias and variance.

Definition: MSE/RMSE

Suppose we have an estimator $\hat{\theta}$ for an estimand θ . Then, we define the mean squared error (MSE) as

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + (\text{bias}(\hat{\theta}))^2$$

The root mean squared error (RMSE) is just the square root of this quantity.

Proof: Let $X = \hat{\theta} - \theta$. Then,

$$\begin{aligned}\text{MSE}(\hat{\theta}) &= \mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2 \\ &= \text{Var}(\hat{\theta} - \theta) + (\mathbb{E}[\hat{\theta}] - \mathbb{E}[\theta])^2 \\ &= \text{Var}(\hat{\theta}) + (\text{bias}(\hat{\theta}))^2\end{aligned}$$

3 Asymptotics

3.1 Consistency of estimators

Often, we'd like to study how an estimator behaves asymptotically, particularly as we increase the sample size $n \rightarrow \infty$. Our hope is usually that the estimator eventually converges to the true value of the estimand.

Definition: Consistency

An estimator $\hat{\theta}$ is consistent for the estimand θ if $\hat{\theta}$ converges in probability to θ as the sample size $n \rightarrow \infty$:

$$\hat{\theta} \xrightarrow{p} \theta$$

More formally, for every $\varepsilon > 0$, we have

$$\mathbb{P}_{\mathbf{Y}}(|\hat{\theta} - \theta| \geq \varepsilon; \theta) \rightarrow 0$$

We also have a nice theorem that gives us a sufficient condition for consistency of an estimator.

Theorem 3.5.4: Sufficient condition for consistency

If $\hat{\theta}$ is an estimator for θ , and $\text{MSE}(\hat{\theta}; \theta) \rightarrow 0$ as $n \rightarrow \infty$, then, $\hat{\theta}$ is consistent.

Proof: Suppose that the MSE of $\hat{\theta}$ converges to 0. Then for any $\varepsilon > 0$,

$$\begin{aligned} \mathbb{P}_{\mathbf{Y}}(|\hat{\theta} - \theta| \geq \varepsilon; \theta) &= \mathbb{P}_{\mathbf{Y}}((\hat{\theta} - \theta)^2 \geq \varepsilon^2; \theta) \\ &\leq \frac{\mathbb{E}[(\hat{\theta} - \theta)^2; \theta]}{\varepsilon^2} \text{ by Markov's inequality} \end{aligned}$$

Note that the numerator $\mathbb{E}[(\hat{\theta} - \theta)^2; \theta] = \text{MSE}(\hat{\theta}; \theta) \rightarrow 0$. Thus, we've shown $\hat{\theta} \xrightarrow{p} \theta$.

In this example, we show that the ECDF is consistent for the true CDF!

Example: Consistency of the ECDF

Suppose we have data $Y_1, \dots, Y_n$ which are distributed i.i.d. under some CDF F . We'll show that the ECDF $\hat{F}$ is consistent.

$$\hat{F}_n(y) = \frac{1}{n} \sum_{j=1}^n \mathbb{1}(Y_j \leq y)$$

The ECDF can be interpreted as a sample mean of indicators, which means it converges in probability to the expected value of the indicator by the LLN.

$$\hat{F}_n(y) \xrightarrow{p} \mathbb{E}[\mathbb{1}(Y_1 \leq y)] = F_{Y_1}(y)$$

3.2 Asymptotic tools

In this section, we review/introduce the five core tools for problems involving asymptotics.

1. Law of large numbers (LLN)
2. Central limit theorem (CLT)
3. Continuous mapping theorem (CMT)
4. Slutsky's theorem
5. Delta method

Definition: Weak law of large numbers (LLN)

Suppose we have i.i.d. $X_1, X_2, X_3 \dots$ with finite mean μ and finite variance. For all positive integers n , let

$$\bar{X}_n = \frac{1}{n} \sum_{j=1}^n X_j$$

Then, the weak LLN says that for all $\epsilon > 0$, $\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \rightarrow 0$, which can be written

$$\bar{X}_n \xrightarrow{p} \mu$$

Definition: Central limit theorem (CLT)

Suppose we have i.i.d. $X_1, X_2, X_3 \dots$ with finite mean μ and finite variance, where $\bar{X}_n$ is the sample mean of $X_1, \dots, X_n$. Then, the CLT states that as $n \rightarrow \infty$,

$$\sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right) \xrightarrow{d} \mathcal{N}(0, 1)$$

Equivalently, we have

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

It follows that an approximate form of the CLT for sufficiently large n is

$$\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$$

Definition: Continuous mapping theorem (CMT)

Let g be a continuous function and let $X_1, X_2, \dots$ be a sequence of r.v.s. If

$$X_n \xrightarrow{p} X,$$

then

$$g(X_n) \xrightarrow{p} g(X).$$

Also, if

$$X_n \xrightarrow{d} X,$$

then

$$g(X_n) \xrightarrow{d} g(X)$$

Definition: Slutsky's theorem

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, where c is a constant, then

$$1) X_n + Y_n \xrightarrow{d} X + c$$

$$2) X_n - Y_n \xrightarrow{d} X - c$$

$$3) X_n Y_n \xrightarrow{d} cX$$

$$4) X_n / Y_n \xrightarrow{d} X / c$$

Note that if $X_n \xrightarrow{d} b$, where b is a constant, then $X_n \xrightarrow{p} b$. This means that if X_n converges to a constant, then we can reapply the same properties above, replacing the convergence in distribution with probability.

As can be seen by the form, Slutsky's theorem naturally pairs well with LLN and CMT, particularly if our estimator can be broken down into smaller terms.

Definition: Delta method

Suppose that g is a differentiable function, and

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Then,

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, (g'(\theta))^2 \sigma^2)$$

As can be seen by the form, the Delta method pairs well with the CLT, especially if an estimator of interest is function of a sample moment.

4 Maximum Likelihood Estimation

4.1 Properties of MLE

We will study in-depth the properties of the MLE. Recall the definition of the MLE:

$$\hat{\theta}_{\text{MLE}} = \underset{\theta \in \Theta}{\operatorname{argmax}} \mathcal{L}(\theta; \mathbf{Y}) = \underset{\theta \in \Theta}{\operatorname{argmax}} \log \mathcal{L}(\theta; \mathbf{Y})$$

Under regularity conditions, which we will define later, the MLE has several useful properties.

Remark: Core properties of the MLE

The MLE $\hat{\theta}$ has the following properties under regularity conditions.

1. Invariance : $\hat{\theta}$ is the MLE of $\theta \Rightarrow g(\hat{\theta})$ is the MLE of $g(\theta)$, if g is 1:1.
2. Consistency : $\hat{\theta}_{\text{MLE}} \xrightarrow{p} \theta$.
3. Asymptotically Normal
4. Asymptotically unbiased
5. Asymptotically efficient : No other estimator has lower variance asymptotically.

We begin by showing invariance, then we will show the remaining properties in later sections.

Theorem: Invariance of the MLE

Suppose $\hat{\theta}$ is the MLE for θ . Then, for a one-to-one transformation g , the MLE of $g(\theta)$ is $g(\hat{\theta})$. This simply follows from invariance of the likelihood function.

4.2 Score, information, efficiency

We define some useful quantities in relation to the likelihood function.

Definition: Score function

The score function is the first derivative of the log-likelihood.

$$s(\theta; \mathbf{Y}) = \frac{\partial \log \mathcal{L}(\theta; \mathbf{Y})}{\partial \theta}$$

Note that the score function, when set to zero, gives us the MLE $\hat{\theta}$. We will see that the score function also allows us to express our uncertainty about $\hat{\theta}$.

Theorem 4.4.3: Information equality

Call θ^* the true value of θ . Under some regularity conditions (most importantly that the support of $\mathbf{Y}$ does not depend on θ , the following hold:

- 1) $\mathbb{E}[s(\theta^*; \mathbf{Y})] = 0$
- 2) $\text{Var}(s'(\theta^*; \mathbf{Y})) = -\mathbb{E}[s'(\theta^*; \mathbf{Y})]$

Next, we introduce Fisher information, which tells us how informative the data are about our parameter θ .

Definition: Fisher information

For a parametric model $f_{\mathbf{Y}}(\mathbf{y}; \theta)$, the Fisher information in $\mathbf{Y}$ about θ is

$$\mathcal{I}_{\mathbf{Y}}(\theta) = \text{Var}(s(\theta; \mathbf{Y})) = \mathbb{E}[s(\theta; \mathbf{Y})^2] = -\mathbb{E}[s'(\theta; \mathbf{Y})]$$

Note that $\mathcal{I}_{\mathbf{Y}}(\theta)$, the sample Fisher information, is different from $\mathcal{I}_{Y_1}(\theta)$, the Fisher information from a single datapoint. If the data are i.i.d., then

$$\mathcal{I}_{\mathbf{Y}}(\theta) = n\mathcal{I}_{Y_1}(\theta)$$

Intuitively, a higher Fisher information and thus a higher variance of the score function means the score function has more distinct peaks, which gives us more “information” on where the most likely location of θ . We can also express the Fisher information understanding a transformation g of the parameter θ .

Theorem 4.4.9: Fisher information under transformation of the parameter

Let $\tau = g(\theta)$, where g is a differentiable function and $g'(\theta) \neq 0$. Then

$$\mathcal{I}_{\mathbf{Y}}(\tau) = \frac{\mathcal{I}_{\mathbf{Y}}(\theta)}{(g'(\theta))^2}$$

Next, we introduce a remarkable theorem about the lower bound of the variance of any unbiased estimator for θ .

Definition: Cramér-Rao lower bound

Let $\hat{\theta} = T(\mathbf{Y})$ be an unbiased estimator for θ for a parametric model $F_{\mathbf{Y}, \theta}$, then the following lower bound holds for all $\hat{\theta}$.

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}_{\mathbf{Y}}(\theta)}$$

Next, we examine the asymptotic distribution of the MLE.

Theorem 4.5.1: Asymptotic distribution of the MLE

Suppose we have the data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} F_{Y;\theta^*}$. Let $\hat{\theta}$ be the MLE for θ , and assume that regularity conditions hold. Then,

$$\sqrt{n}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_{Y_1}^{-1}(\theta^*))$$

4.3 Regularity conditions

Many of the useful properties of the MLE hold only under a certain set of conditions. Now, let's clearly define our regularity conditions, and see what happens to the MLE whenever they do not hold!

Definition: Regularity conditions

We define the following regularity conditions for the standard, general properties of the MLE to apply.

1. Support of data not dependent on θ (ex: $\text{Unif}(0, \theta)$ data)
2. Identifiability : If $\theta_1 \neq \theta_2$, then $F(y; \theta_1) \neq F(y; \theta_2)$.
 - Ex: $Y_j \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\alpha + \beta, \sigma^2)$, then α and β cannot be definitively “disentangled” for estimation.
3. DuThIS applies (necessary for $\mathbb{E}[\text{score}]$).
4. True θ^* not on the boundary of parameter space Θ .
5. Third derivative of $\log \mathcal{L}(\theta)$ exists (smoothness).

So what happens when regularity conditions do not apply? We will review the famous German tank problem.

Example: German tank problem

Suppose that the Allies capture n German tanks, where our estimand θ is the total number of German tanks in existence. The tanks are numbered $1, 2, \dots, \theta$. Our data is $y_1, \dots, y_n$, where y_j is the serial number of the j th captured tanks. We will find the MLE of θ .

Note that our regularity conditions don't apply, since the support of our data is dependent on our estimand θ .

We begin with the likelihood function, which we derive using counting.

$$\mathcal{L}(\theta; \mathbf{Y}) = \begin{cases} \frac{1}{\binom{\theta}{n}} & y_j \leq \theta \\ 0 & \text{otherwise} \end{cases} = \frac{\mathbb{1}(y_{(n)} \leq \theta)}{\binom{\theta}{n}}$$

For the likelihood function to be nonzero, the largest serial number $y_{(n)}$ must be less than the estimand θ . So, if we plot $\mathcal{L}(\theta)$ vs. θ , we will see that the likelihood begins at 0, spikes at $y_{(n)}$, then decreases as θ increases. Thus,

$$\hat{\theta}_{\text{MLE}} = y_{(n)}$$

Notice that this is nonsensical! It means our “best” guess of the total number of German tanks is just the highest serial number in our sample!

5 Confidence Intervals

5.1 Interval estimators

So far, we've been looking at point estimators for an estimand θ . But what if we want to give ourselves a little bit more leeway in our estimate? We can do exactly this by finding confidence intervals.

Definition: Interval estimator/estimate

Suppose we collect data $\mathbf{y}$, and we specify a model with parameter θ . Then, the following interval is an interval estimate for θ .

$$C(\mathbf{y}) = [L(\mathbf{y}), U(\mathbf{y})]$$

Note that both terms are purely functions of the data! The corresponding random version is an interval estimator:

$$C(\mathbf{Y}) = [L(\mathbf{Y}), U(\mathbf{Y})]$$

But how do we actually evaluate how often our interval captures the true, fixed (in the frequentist view) value of θ ?

Definition: Coverage probability

Suppose the interval estimator for θ is $C(\mathbf{Y})$. We say that an interval $C(\mathbf{y})$ covers θ if $\theta \in C(\mathbf{y})$. Then we define the coverage probability of the interval estimator $C(\mathbf{Y})$ as

$$P_{\mathbf{Y}}(\theta \in C(\mathbf{Y}); \theta) = \int \mathbb{1}(\theta \in C(\mathbf{y})) f_{\mathbf{Y}}(\mathbf{y}; \theta) d\mathbf{y}$$

In other words, this is the expected fraction of times that our random interval $C(\mathbf{Y})$ covers the fixed, true value of θ . Note that the coverage probability is a function of θ !

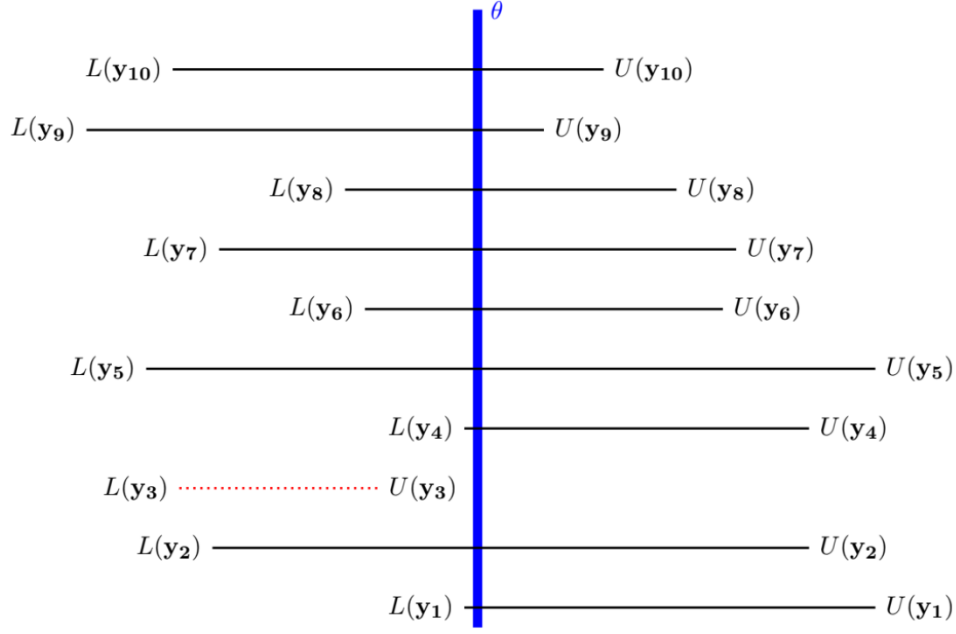


Figure 1: Illustration of the coverage probability of the interval $C(\mathbf{Y}) = [L(\mathbf{Y}), U(\mathbf{Y})]$. Each crystallization of $\mathbf{Y} = \mathbf{y}_j$ yields a different interval $[L(\mathbf{y}_j), U(\mathbf{y}_j)]$, and the coverage probability tells us what percent of the time our random interval crystallizes “around” the true θ .

The typical procedure to construct an interval estimator is to begin by pre-setting the desired coverage probability, then finding the corresponding functions of the data $L(\mathbf{Y})$ and $U(\mathbf{Y})$ that give us an interval $[L(\mathbf{Y}), U(\mathbf{Y})]$ with that coverage probability!

Definition: Confidence interval

Fix a number $\alpha \in (0, 1)$. Then, the interval estimator $C(\mathbf{Y}) = [L(\mathbf{Y}), U(\mathbf{Y})]$ is a (1- α) confidence interval if

$$\mathbb{P}(L(\mathbf{Y}) \leq \theta \leq U(\mathbf{Y})) = 1 - \alpha$$

The number $1 - \alpha$ is also called the level.

5.2 Constructing confidence intervals

Note that θ is viewed as a fixed, unknown quantity, so we cannot speak about its distribution in the frequentist framework. To construct a confidence interval, we will need to define quantities with known distributions, so that we are able to make actual probability statements about. This will allow us to ensure that our interval actually attains the $1 - \alpha$ confidence level we want. We define a pivot below.

Definition: Pivot

A pivot is a random variable with a known distribution.

So what does this look like in practice?

Example: Pivot for i.i.d. Normal data

Suppose we have data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where σ^2 is known. Then, by convolution of independent Normals, the sample mean is distributed

$$\bar{Y} \sim \mathcal{N}(\mu, \sigma^2/n)$$

Then, by location-scale,

$$\frac{\sqrt{n}(\bar{Y} - \mu)}{\sigma} \sim \mathcal{N}(0, 1)$$

This is a pivot!

Let's do an example of constructing a CI to make this more concrete.

Example: Standard normal pivot

We'd like to construct a 95% CI for an estimand θ . Suppose we have an estimator $\hat{\theta}$ for θ , where

$$\hat{\theta} \sim \mathcal{N}(\theta, \sigma^2),$$

where σ^2 is known. In practice, such a result would often come from an asymptotic approximation, which we explore in the next section. Then, we obtain the pivot

$$\frac{\hat{\theta} - \theta}{\sigma} \sim \mathcal{N}(0, 1)$$

Now, we can simply trap our pivot between an upper and lower bound.

$$\mathbb{P}\left(a \leq \frac{\hat{\theta} - \theta}{\sigma} \leq b\right) = \Phi(b) - \Phi(a) = 0.95$$

So, an intuitive choice for the upper and lower bound would be to use the $\mathcal{N}(0, 1)$ quantile function, defined $Q_{\mathcal{N}(0,1)}$!

$$a = Q_{\mathcal{N}(0,1)}(0.025) \approx -1.96, \quad b = Q_{\mathcal{N}(0,1)}(0.975) \approx 1.96$$

Then, we can rearrange the terms inside of the probability:

$$\mathbb{P}\left(-1.96 \leq \frac{\hat{\theta} - \theta}{\sigma} \leq 1.96\right) = \mathbb{P}(\hat{\theta} - 1.96\sigma \leq \theta \leq \hat{\theta} + 1.96\sigma) = 0.95$$

Thus, we obtain a 95% CI for θ :

$$[\hat{\theta} - 1.96\sigma, \hat{\theta} + 1.96\sigma]$$

Above we've derived an incredibly useful result whenever we obtain a Normal pivot.

Remark: $(1 - \alpha)$ CI using Normal pivot

Suppose we have an estimand $\hat{\theta}$ for θ with known variance σ^2 , such that

$$\hat{\theta} \sim \mathcal{N}(\theta, \sigma^2) \Rightarrow \frac{\hat{\theta} - \theta}{\sigma} \sim \mathcal{N}(0, 1)$$

Then, we can write

$$\mathbb{P} \left(Q_{\mathcal{N}(0,1)}(\alpha/2) \leq \frac{\hat{\theta} - \theta}{\sigma} \leq Q_{\mathcal{N}(0,1)}(1 - \alpha/2) \right) = 1 - \alpha$$

Thus, rearranging to isolate θ in the inequality, the $(1 - \alpha)$ CI is

$$[\hat{\theta} + \sigma Q_{\mathcal{N}(0,1)}(\alpha/2), \hat{\theta} + \sigma Q_{\mathcal{N}(0,1)}(1 - \alpha/2)]$$

For $\alpha = 0.05$ (95% CI), this simply becomes

$$\hat{\theta} \pm 1.96\text{SE}(\hat{\theta})$$

Visually, we intuit this result using the tails of the Normal.

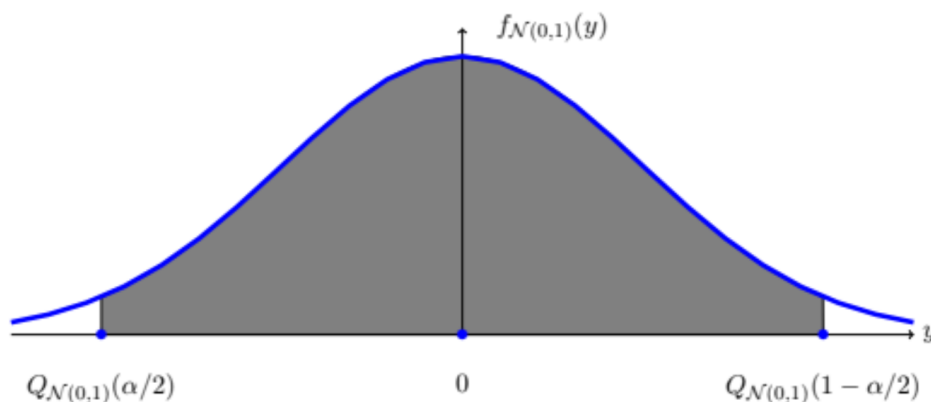


Figure 2: $1 - \alpha$ CI based on $\mathcal{N}(0, 1)$, where each tail has $\alpha/2$ area such that the shaded region contains $1 - \alpha$ probability.

So, the general workflow for confidence intervals is...

1. Construct pivot with known distribution.

2. Make probability statement about where that pivot falls using quantiles.
3. Isolate θ in the inequality to obtain a CI.

5.3 Asymptotic approximations

In practice, it's not common for our estimator $\hat{\theta}$ to have nice distributional properties that allow us to immediately construct a pivot. In these cases, we must rely on the asymptotic distribution of our estimator $\hat{\theta}$ in order to construct approximate pivots.

Definition: Approximate pivot

Suppose that we an estimator $\hat{\theta}$ such that

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Then, for sufficiently large n ,

$$\hat{\theta} \sim \mathcal{N}(\theta, \sigma^2/n)$$

Thus, we derive the approximate pivot

$$\frac{\sqrt{n}(\hat{\theta} - \theta)}{\sigma} \sim \mathcal{N}(0, 1)$$

An approximate pivot can be derived using asymptotic properties of an estimator, which often follow from CLT and/or the delta method.

6 Regression

6.1 Regression frameworks

There are two different frameworks for regression. Both involve data that come in pairs $(\mathbf{X}, \mathbf{Y}) = (X_1, Y_1), \dots, (X_n, Y_n)$.

1. Conditional learning of $\mathbb{E}[Y|(X = x)] = \mu(x)$. In this framework, we use X to predict Y using the conditional distribution of $Y|X = x$. So, this framework is often denoted predictive regression.
2. Unconditional learning of $g(X)$ such that $Y - g(X)$ is small, on average, over the joint distribution of X, Y . In this framework, we are simply finding a transformation of X that pushes it as close as possible to Y , for example, by minimizing the expected squared distance.

6.2 Conditional learning: linear regression

One central goal of statistics is to predict the outcome Y given a set of predictors $(X_1, \dots, X_n)$. We define predictive regression more formally below:

Definition: Predictive regression

The task of estimating

$$\mu(x) = \mathbb{E}[Y|\mathbf{X} = \mathbf{x}]$$

is known as predictive regression. Here, Y is an outcome or label, and the $\mathbf{X}$ are called predictors, covariates, or features.

It is also critical to be clear about our assumptions on the conditional variance of $Y|\mathbf{X} = \mathbf{x}$.

Definition: Hetero and homoskedasticity

Write

$$\sigma^2(x) = \text{Var}(Y|\mathbf{X} = \mathbf{x}).$$

If $\sigma^2(x)$ does not vary with x , then the regression is homoskedastic. Otherwise, it is heteroskedastic.

We also define some quantities that deal with how close the prediction $\mathbb{E}[Y|\mathbf{X} = \mathbf{x}]$ is to the true value of Y .

Definition: Regression error

We define the regression error as the random variable

$$U(\mathbf{x}) = Y - \mathbb{E}[Y|\mathbf{X} = \mathbf{x}]$$

Note that by construction (Adam's law),

$$\mathbb{E}[U(\mathbf{X})|\mathbf{X} = \mathbf{x}] = 0$$

Definition: Signal-noise decomposition

We can decompose the outcome Y into a regression term (signal) and an error term (noise).

$$Y = \mu(\mathbf{x}) + U(\mathbf{x})$$

A popular choice of regression model is linear regression, which we define below.

Definition: Linear regression

A linear regression model for the outcome Y given the predictors $\mathbf{X} = (X_1, \dots, X_n)$ can be written as

$$\mathbb{E}[Y|\mathbf{X} = \mathbf{x}] = \theta_0 + \theta_1 x_1 + \dots + \theta_K x_K,$$

where $\theta = (\theta_0, \theta_1, \dots, \theta_K)^T$. The elements of θ are called the regression coefficients, where θ_0 is the intercept and the remaining θ_j are the slopes.

Note that linear regression is linear in the parameters, not in the X_j ! So, for example,

$$\mathbb{E}[Y|\mathbf{X} = \mathbf{x}] = \theta_0 + \theta_1 x_1 + \theta_2 x_1 x_2$$

would also be a valid linear regression model, where we include an interaction term $\theta_2 x_1 x_2$. Note that such a model holds for all $\mathbf{x}$, but it can be dangerous to extrapolate outside of the range of the data $\mathbf{X} = \mathbf{x}$.

6.3 Conditional learning: logistic regression

Next, we introduce logistic regression, which is used when the outcome Y is binary: $Y \in \{0, 1\}$. In this case, we can write our regression as

$$\mu(\mathbf{x}) = \mathbb{E}[Y|\mathbf{X} = \mathbf{x}] = \mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x}) \text{ by fund. bridge}$$

We now define the odds and log-odds, which will prove useful for our logistic regression model.

Definition: Logit function

The logit function is defined as

$$\text{logit}(p) = \log \left(\frac{p}{1-p} \right).$$

The inverse logit/logistic/sigmoid function is

$$\text{logit}^{-1}(x) = \frac{e^x}{1 + e^x}$$

Definition: Odds and log-odds in logistic regression

Recall the quantity

$$\mu(\mathbf{x}) = \mathbb{E}[Y|\mathbf{X} = \mathbf{x}] = \mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x})$$

Then, we define the odds as

$$\frac{\mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x})}{\mathbb{P}(Y = 0|\mathbf{X} = \mathbf{x})} = \frac{\mu(\mathbf{x})}{1 - \mu(\mathbf{x})}$$

and the log-odds as

$$\lambda(\mathbf{x}) = \log \left\{ \frac{\mu(\mathbf{x})}{1 - \mu(\mathbf{x})} \right\} = \log \mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x}) - \log \mathbb{P}(Y = 0|\mathbf{X} = \mathbf{x})$$

Note here that we can relate the probability of success $\mu(\mathbf{x})$ and the log-odds $\lambda(\mathbf{x})$ as follows:

$$\begin{aligned} 1) \lambda(\mathbf{x}) &= \text{logit}(\mu(\mathbf{x})) = \log \left\{ \frac{\mu(\mathbf{x})}{1 - \mu(\mathbf{x})} \right\} \\ 2) \mu(\mathbf{x}) &= \text{logit}^{-1}(\lambda(\mathbf{x})) = \frac{e^{\lambda(\mathbf{x})}}{1 + e^{\lambda(\mathbf{x})}} \end{aligned}$$

Now, we formally define a logistic regression model.

Definition: Logistic regression

For a pair of data $(\mathbf{X}, Y)$, where $Y \in \{0, 1\}$, define the quantity

$$\mu(\mathbf{x}) = \mathbb{E}[Y|\mathbf{X} = \mathbf{x}] = \mathbb{P}(Y = 1|\mathbf{X} = \mathbf{x})$$

Then, logistic regression models the log-odds $\lambda(\mathbf{x})$ defined above as linear in param-

ters:

$$\lambda(\mathbf{x}) = \log \left\{ \frac{\mu(\mathbf{x})}{1 - \mu(\mathbf{x})} \right\} = \theta_0 + \theta_1 x_1 + \cdots + \theta_K x_K,$$

which means we can write the probability of success $\mu(\mathbf{x})$ as

$$\mu(\mathbf{x}) = \frac{e^{\theta_0 + \theta_1 x_1 + \cdots + \theta_K x_K}}{1 + e^{\theta_0 + \theta_1 x_1 + \cdots + \theta_K x_K}}$$

6.4 Statistical models of predictive regression

Suppose we have the data

$$(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_j, y_j).$$

Then, a common approach is to impose a statistical model on the conditional distribution of the outcome given the predictors by specifying

$$\mathbf{Y}|\mathbf{X} = \mathbf{x}; \theta$$

Under this framework, we can perform inference using the exact same procedures we've already seen in previous chapters, except we've now conditioned everything on $\mathbf{X} = \mathbf{x}$.

Remar: Conditional likelihood for predictive regression

If we assume conditional independence of the outcomes $Y_{1:n}$ given the predictors $\mathbf{X}_{1:n} = \mathbf{x}_{1:n}$, we can write the conditional likelihood as

$$\begin{aligned} \mathcal{L}(\theta; y_{1:n} | \mathbf{X}_{1:n} = \mathbf{x}_{1:n}) &= f(y_{1:n} | \mathbf{X}_{1:n} = \mathbf{x}_{1:n}; \theta) \\ &= \prod_{j=1}^n f(y_j | \mathbf{X}_j = \mathbf{x}_j; \theta) \end{aligned}$$

and the log-likelihood as

$$\log \mathcal{L}(\theta; y_{1:n} | \mathbf{X}_{1:n} = \mathbf{x}_{1:n}) = \sum_{j=1}^n \log f(y_j | \mathbf{X}_j = \mathbf{x}_j; \theta)$$

So, we can directly apply everything we derived when dealing with $\mathcal{L}(\theta; \mathbf{Y})$ in previous units, except now everything is conditioned on $\mathbf{X}_{1:n} = \mathbf{x}_{1:n}$!

- 1) $\hat{\theta}_{\text{MLE}} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log \mathcal{L}(\theta; y_{1:n} | \mathbf{X}_{1:n}, \mathbf{x}_{1:n})$
- 2) $\mathbb{E}[s(\theta; y_{1:n} | \mathbf{X}_{1:n}, \mathbf{x}_{1:n})] = 0$
- 3) $\mathcal{I}_{\mathbf{Y}|\mathbf{X}=\mathbf{x}}(\theta) = \operatorname{Var}(s(\theta; y_{1:n} | \mathbf{X}_{1:n}, \mathbf{x}_{1:n})) = -\mathbb{E}[s'(\theta; y_{1:n} | \mathbf{X}_{1:n}, \mathbf{x}_{1:n})]$

Next, let's examine the case of Gaussian linear regression, which assumes the scatter around $\mathbb{E}[Y|\mathbf{X} = \mathbf{x}]$ is distributed Gaussian.

Example: Gaussian linear regression with no intercept

Impose a Gaussian linear regression model on outcomes $\mathbf{Y} = (Y_1, \dots, Y_n)$ and predictors $\mathbf{X} = (X_1, \dots, X_n)$ such that

$$Y_j | \mathbf{X} = \mathbf{x} \stackrel{\text{ind.}}{\sim} \mathcal{N}(\theta x_j, \sigma^2)$$

We would like to estimate the slope θ . The corresponding log-likelihood, score, and Fisher information for this model are

$$1) \log \mathcal{L}(\theta) = -\frac{1}{2\sigma^2} \sum_{j=1}^n (y_j - \theta x_j)^2$$

$$2) s(\theta) = \frac{1}{\sigma^2} \sum_{j=1}^n x_j (y_j - \theta x_j)$$

$$3) \mathcal{I}_{Y|\mathbf{X}=\mathbf{x}}(\theta) = \frac{1}{\sigma^2} \sum_{j=1}^n x_j^2$$

Then, the MLE for θ is

$$\hat{\theta} = \frac{\sum_{j=1}^n x_j y_j}{\sum_{j=1}^n x_j^2}$$

This estimator is also known as the ordinary least-squares (OLS) estimator. We will study its properties in depth in the following section.

Next, we examine the residuals, which are the observed prediction errors based on our data and estimated $\hat{\theta}$.

Definition: Residuals

Suppose we have estimated θ using $\hat{\theta}$. Then, it is natural to predict the Y_j using $\hat{\theta}x_j$, the predicted/fitted value of Y_j . Then, the residuals are defined

$$\hat{U}_j = Y_j - \hat{\theta}x_j,$$

the difference between the true and predicted outcomes. Note that the residuals are observed, whereas the true errors U_j are unknown, hence the hat in the notation for residuals.

6.5 Ordinary least-squares estimator

We will closely examine the properties of the OLS estimator, which we redefine below

Definition: OLS estimator

Suppose we have outcomes $\mathbf{Y} = (Y_1, \dots, Y_n)$, predictors $\mathbf{X} = (X_1, \dots, X_n)$. We assume conditional independence of the Y_j given the predictors $\mathbf{X} = \mathbf{x}$ and specify a regression model

$$\mathbb{E}[Y_j | \mathbf{X} = \mathbf{x}] = \theta x_j$$

Then, the OLS estimator is defined

$$\hat{\theta}_{\text{OLS}} = \frac{\sum_{j=1}^n x_j Y_j}{\sum_{j=1}^n x_j^2}$$

We introduce some core properties of this estimator below.

Theorem 6.3.6: Properties of the OLS estimator

Suppose we have data $\mathbf{Y}, \mathbf{X}$, where the Y_j are conditionally independent given the predictors $\mathbf{X} = \mathbf{x}$. We use $\hat{\theta}$, the OLS estimator. We define

$$\mu(x_j) = \mathbb{E}[Y_j | X_j = x_j], \sigma_j^2 = \text{Var}(Y_j | X_j = x_j).$$

Then,

$$\begin{aligned} \mathbb{E}[\hat{\theta} | \mathbf{X} = \mathbf{x}] &= \left(\sum_{j=1}^n x_j^2 \right)^{-1} \left(\sum_{j=1}^n x_j \mu(x_j) \right), \\ \text{Var}(\hat{\theta} | \mathbf{X} = \mathbf{x}) &= \left(\sum_{j=1}^n x_j^2 \right)^{-2} \left(\sum_{j=1}^n x_j^2 \sigma_j^2 \right) \end{aligned}$$

Proof: For the expectation, we simply apply linearity.

$$\mathbb{E}[\hat{\theta} | \mathbf{X} = \mathbf{x}] = \left(\sum_{j=1}^n x_j^2 \right)^{-1} \left(\sum_{j=1}^n x_j \mathbb{E}[Y_j | \mathbf{X} = \mathbf{x}] \right) = \left(\sum_{j=1}^n x_j^2 \right)^{-1} \left(\sum_{j=1}^n x_j \mu(x_j) \right)$$

For the variance, we apply conditional independence to sum the variances.

$$\text{Var}(\hat{\theta} | \mathbf{X} = \mathbf{x}) = \left(\sum_{j=1}^n x_j^2 \right)^{-2} \left(\sum_{j=1}^n x_j^2 \text{Var}(Y_j | \mathbf{X} = \mathbf{x}) \right) = \left(\sum_{j=1}^n x_j^2 \right)^{-2} \left(\sum_{j=1}^n x_j^2 \sigma_j^2 \right)$$

Next, we apply the theorem above to derive a lemma that gives the properties of the OLS estimator.

mator under successively stronger assumptions. We end with the Gaussian linear regression model.

Lemma 6.3.7: OLS estimator properties under three assumptions

(a) If $\mu_j = \theta x_j$, then

$$\mathbb{E}[\hat{\theta}|\mathbf{X} = \mathbf{x}] = \theta,$$

meaning the OLS estimator is conditionally unbiased given the predictors.

(b) If homoskedasticity holds, meaning $\sigma_j^2 = \sigma^2$, then

$$\text{Var}(\hat{\theta}|\mathbf{X} = \mathbf{x}) = \sigma^2 / \sum_{j=1}^n x_j^2$$

(c) If we assume a Gaussian linear regression model such that

$$Y_j|\mathbf{X} = \mathbf{x} \sim \mathcal{N}(\theta X_j, \sigma^2),$$

then $\hat{\theta}$ is conditionally unbiased for θ and conditionally achieves the CRLB

$$\mathbb{E}[\hat{\theta}|\mathbf{X} = \mathbf{x}] = \theta, \text{Var}(\hat{\theta}|\mathbf{X} = \mathbf{x}) = \mathcal{I}_{\mathbf{Y}|\mathbf{X}=\mathbf{x}}^{-1}(\theta)$$

Furthermore,

$$\hat{\theta}|\mathbf{X} = \mathbf{x} \sim \mathcal{N}\left(\theta, \sigma^2 / \sum_{j=1}^n x_j^2\right)$$

Note above that we actually do not need homoskedasticity for the OLS estimator to be conditionally unbiased! All we need is that $\mu_j = \theta x_j$. So what happens if we allow heteroskedasticity such that σ_j^2 varies with j ? We introduce the robust standard error.

Definition: Robust standard error

Suppose that condition (a) from Lemma 6.3.7 holds, but we allow heteroskedasticity such that σ_j^2 varies with j . Recall that the conditional variance of the OLS estimator can be written

$$\text{Var}(\hat{\theta}|\mathbf{X} = \mathbf{x}) = \left(\sum_{j=1}^n x_j^2\right)^{-2} \left(\sum_{j=1}^n x_j^2 \sigma_j^2\right).$$

Then, the heteroskedastically robust variance estimator is

$$\widehat{\text{Var}}(\hat{\theta}|\mathbf{X} = \mathbf{x}) = \left(\sum_{j=1}^n x_j^2\right)^{-2} \left(\sum_{j=1}^n x_j^2 \hat{U}_j^2\right)$$

The square root of this estimator is the robust standard error. These estimators estimate the conditional variance of the OLS estimator even when homoskedasticity (which is often a strong assumption) does not hold.

6.6 Unconditional learning: linear projection

Previously, we examined properties of the conditional distribution $Y|X = x$, such as

$$\mu(x) = \mathbb{E}[Y|X = x], \sigma^2(x) = \text{Var}(Y|X = x)$$

Now, we introduce a completely different framework. Instead of dealing with the conditional distribution of $Y|X = x$, we now focus on the joint distribution of (X, Y) . Note that there is no directional assumption in the unconditional approach. We are looking at the “joint movement” of the X and Y . Some familiar measures of the joint distribution are

$$\mathbb{E}[X], \mathbb{E}[Y], \text{Var}(X), \text{Var}(Y), \text{Cov}(X, Y), \text{Corr}(X, Y)$$

The idea will be to find a function g such that

$$Y - g(X) \text{ is small, averaged over } X, Y$$

A special case is where the function $g(X)$ is linear in X .

Definition: Linear projection

Within the unconditional learning framework, we'd like to minimize $Y - g(X)$. if $g(X)$ is linear in X , then we define g using the linear projection LP:

$$\text{LP}(Y|X) = \alpha + \theta X$$

7 Exponential Families and Data Compression

7.1 Sufficient statistics

It's often the case that we have a compressed form $T(\mathbf{Y})$ of the data we want to examine $\mathbf{Y}$. In these cases, the question arises, "Can we still estimate a parameter θ even without seeing the full data $\mathbf{Y}$?" This is where the notion of a sufficient statistics arises.

Definition: Sufficient statistic

Suppose we have data $\mathbf{Y}$ and a parametric model $F_{\mathbf{Y};\theta}$. A statistic $T(\mathbf{Y})$ is sufficient if the distribution of $\mathbf{Y}|T(\mathbf{Y})$ is not a function of θ . Intuitively, this means that once we condition on the sufficient statistic, there is no more "information" left about θ in $\mathbf{Y}$. The sufficient statistic gives us everything we need to know about θ !

We define the following criterion to assess whether a statistic is sufficient.

Theorem 7.2.5 in textbook: Factorization criterion

Suppose we wish to estimate θ given a model $F_{\mathbf{Y};\theta}$ using a statistic $T(\mathbf{Y})$. The statistic $T(\mathbf{Y})$ is sufficient if and only if we can factor

$$f_{\mathbf{Y}}(\mathbf{y}; \theta) = g_{\theta}(t)h(\mathbf{y})$$

Above, $g_{\theta}(t)$ is a function of the computed value of the test statistic, and $h(\mathbf{y})$ is purely a function of the data, and not θ . For some intuition, note that if the factorization criterion holds, then

$$\mathcal{L}(\theta; \mathbf{Y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta) = g_{\theta}(t)h(\mathbf{y}) \propto g_{\theta}(t)$$

This means that $g_{\theta}(t)$ can just be used as the likelihood, since the only difference is a multiplicative constant!

Let's do a few examples to understand how this criterion works.

Example: Sufficient statistic for Poisson data

Suppose we have data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Pois}(\theta)$. We want to find a sufficient statistic to

estimate θ . We can write the joint density

$$\begin{aligned} f_{\mathbf{Y}}(\mathbf{y}; \theta) &= \prod_{j=1}^n \frac{e^{-\theta} \theta^{y_j}}{y_j!} \text{ by ind., Poisson PMF} \\ &= e^{-n\theta} \cdot \theta^{n\bar{y}} \cdot \prod_{j=1}^n \frac{1}{y_j!} \\ &= g_{\theta}(t) h(\mathbf{y}), \text{ where } g_{\theta}(t) = e^{-n\theta} \theta^{n\bar{y}}, h(\mathbf{y}) = \prod_{j=1}^n \frac{1}{y_j!} \end{aligned}$$

Thus, a sufficient statistic is

$$t = n\bar{y} = \sum_{j=1}^n y_j$$

Note that in general, if T is sufficient, then $g(T)$ is also sufficient, where g is some 1-to-1 function. Sufficient statistics are not unique. We can see in the above example that using $\bar{Y}$ instead of $n\bar{Y}$ would also work to estimate θ !

7.2 Rao-Blackwellization

Now, we introduce a general recipe to improve the variance of an estimator $\hat{\theta}$ given that we have a sufficient statistic T for our data $\mathbf{Y}$.

Theorem 7.2.12 in textbook: Rao-Blackwell

Let $\hat{\theta}$ be some estimator of θ , and let T be a sufficient statistic. Then, the Rao-blackwellized estimator

$$\hat{\theta}_{\text{RB}} = \mathbb{E}[\hat{\theta}|T]$$

has the same bias as $\hat{\theta}$ and variance equal to or less than $\hat{\theta}$:

$$\text{bias}(\hat{\theta}_{\text{RB}}) = \text{bias}(\hat{\theta}), \text{Var}(\hat{\theta}_{\text{RB}}) = \text{Var}(\hat{\theta})$$

Proof:

- 1) $\mathbb{E}[\hat{\theta}_{\text{RB}}] = \mathbb{E}[\mathbb{E}[\hat{\theta}|T]] = \mathbb{E}[\hat{\theta}]$ by Adam's law
- 2) $\text{Var}(\hat{\theta}) = \mathbb{E}[\text{Var}(\hat{\theta}|T)] + \text{Var}(\mathbb{E}[\hat{\theta}|T])$ by Eve's law

$$\begin{aligned} &= \mathbb{E}[\text{Var}(\hat{\theta}|T)] + \text{Var}(\hat{\theta}_{\text{RB}}) \\ &\geq \text{Var}(\hat{\theta}_{\text{RB}}) \text{ by non-negativity of variance} \end{aligned}$$

Note that Rao-Blackwell will not improve an estimator $\hat{\theta} = g(T)$: a the estimator is a function

of a sufficient statistic. This is because by taking out what's known,

$$\mathbb{E}[\hat{\theta}|T] = \mathbb{E}[g(T)|T] = g(T) = \hat{\theta}$$

Let's construct a stupid estimator, then improve it using Rao-Blackwellization!

Example: Improving a dumb estimator (Poisson data)

Suppose we have data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Pois}(\theta)$, we know $n\bar{Y}$ is sufficient, and we'd like to estimate θ . A stupid estimator uses only the first datapoint:

$$\hat{\theta}_{\text{stupid}} = Y_1$$

Let's apply Rao-Blackwell:

$$\hat{\theta}_{\text{RB}} = \mathbb{E}[Y_1 | n\bar{Y}]$$

We find the conditional distribution of Y_1 :

$$Y_1 | (Y_1 + \dots + Y_n) \sim \text{Bin} \left(\sum_{j=1}^n Y_j, \frac{1}{n} \right) \text{ by chicken-egg}$$

Thus, our Rao-Blackwellized estimator is

$$\begin{aligned} \hat{\theta}_{\text{RB}} = \mathbb{E}[Y_1 | (Y_1 + \dots + Y_n)] &= \frac{1}{n} \sum_{j=1}^n Y_j \text{ by Binomial mean} \\ &= \bar{Y} \end{aligned}$$

Rao-Blackwellization has given us the sensible estimator $\hat{\theta}_{\text{RB}} = \bar{Y}$, which turns out to also be $\hat{\theta}_{\text{MLE}}$.

As we see above, a common prerequisite to using Rao-Blackwell is that we know the conditional distribution of the data given the sufficient statistic: $Y_j | T$.

7.3 NEFs and EFs

Next, we cover exponential families, a general form for a distribution that encompasses many of the common distributions we've seen so far. We begin with NEFs.

Definition: Natural exponential family

A density function $f_Y(y; \theta)$ follows a natural exponential family (NEF) if it can be written

$$f_Y(y; \theta) = e^{\theta y - \psi(\theta)} h(y),$$

where $h(y)$ is θ -free. θ is called the natural parameter and may be a reparameterization of the parameters in the originally specified model.

Examples of NEFs include the Binomial, Poisson, Gamma (and therefore Exponential), Negative Binomial, and Normal. Let's examine some global properties that hold for all NEFs.

Remark: Useful properties of NEFs

Suppose Y follows an NEF, meaning $f_Y(y; \theta) = e^{-\theta y - \psi(\theta)} h(y)$. Then,

- 1) $\log \mathcal{L}(\theta; Y) = \theta y - \psi(\theta)$
- 2) $s(\theta; Y) = Y - \psi'(\theta)$
- 3) $\mathbb{E}[Y] = \psi'(\theta)$ from above
- 4) $\mathcal{I}_Y(\theta) = \text{Var}(Y) = \psi''(\theta)$

The above are general properties of NEFs. However, we get even more useful properties when we use an NEF as a statistical model.

Remark: NEFs as statistical models

Suppose we have i.i.d. data $Y_1, \dots, Y_n$ from an NEF with density

$$f_Y(y; \theta) = e^{\theta y - \psi(\theta)} h(y)$$

Let $\mu = \mathbb{E}[Y_1; \theta]$. Then, the following statements are hold:

- 1) $\bar{Y}$ is a sufficient statistic for θ
- 2) $\hat{\mu}_{\text{MLE}} = \bar{Y}$

These results are incredibly useful when we're working with an NEF model! We automatically get the MLE for the mean, and we can apply invariance to get the MLE for any other parameter that is a function of the mean.

We also define the closely related exponential family below.

Definition: Exponential family

Suppose Y is an NEF. Then, $g(Y)$ is an EF. More formally, the density of an EF can be written

$$f_Y(y; \theta) = e^{\theta T(y) - \psi(\theta)} h(y).$$

where h does not depend on θ , and $T(y)$ is some function of the data.

8 Hypothesis testing

8.1 Fundamentals of hypothesis testing

We'll now cover hypothesis testing. Say we've collected data $\mathbf{Y} = \mathbf{y}$, and our data is generated by $F_{Y;\theta}$. Then we define a statistical hypothesis below:

Definition: Statistical hypothesis

We partition the parameter Θ into two disjoint sets Θ_0 and Θ_1 , such that $\Theta_0 \cup \Theta_1 = \Theta$. Then, we define the null (H_0) and alternative (H_1) hypotheses below:

$$H_0 : \theta \in \Theta_0, H_1 : \theta \in \Theta_1$$

We call a hypothesis simple if it is simply a point (ex: $\Theta_0 = \{\theta_0\}$), and composite otherwise.

We want a procedure to carry out to test whether we can reject or retain a hypothesis based on the data $\mathbf{y}$. Thus, we define a procedure known as a hypothesis test below.

Definition: Statistical hypothesis test

We define a procedure to either reject or retain H_0 based on whether the data $\mathbf{y}$ falls within a specified region.

$$\begin{aligned} \text{reject } H_0 : \mathbf{y} \in A^c, & \text{ the } \underline{\text{rejection region}} \\ \text{retain } H_0 : \mathbf{y} \in A, & \text{ the } \underline{\text{retention region}} \end{aligned}$$

Often, it's easier to base our test on a test-statistic $T(\mathbf{y})$ computed from our data. In this case, we define a critical region D such that the following is true.

$$\text{reject } H_0 : T(\mathbf{y}) \in D$$

The set D can take multiple forms such as $D = \{T(\mathbf{y}) > c\}$ or $\{T(\mathbf{y}) : c_L < T(\mathbf{y}) < c_U\}$, where c, c_L, c_U are the critical values.

As we see above, we like to work with a test-statistic computed from the data, then see whether the test-statistic falls within a certain region. After we've performed the hypothesis test, we either reject or retain the null. Note that a hypothesis test compresses our data into a binary outcome. We don't know the truth, so it's possible that we make errors in our binary decision, which are described below.

Definition: Errors in hypothesis testing

- (1) Type I error: We reject H_0 , but H_0 is true ("false positive").
- (2) Type II error: We retain H_0 , but H_1 is true ("false negative").

We want to be able to evaluate our hypothesis tests in terms of the probability that we make the kinds of errors specified above. This will tell how “good” our test is, and how likely it is that we made the wrong judgement call. We define the power function below.

Definition: Power

Assume the data were generated $F_{Y;\theta}$, and we have our hypothesis and retention region A . The power function of the test is

$$\beta(\theta) = \mathbb{P}_Y(Y \notin A; \theta) = \int \mathbb{1}(y \in A^c) f_Y(y; \theta) dy$$

Intuitively, we can think of $\beta(\theta)$ as the “probability of rejecting the null under a specified value of θ .”

Lastly, we’ll define the size of a test, which will also help us evaluate a given testing procedure.

Definition: Size

The size α of a testing procedure is defined as follows:

$$\alpha = \max_{\theta \in \Theta_0} \beta(\theta)$$

Intuitively, we can think of α , as the the “maximal probability of rejecting the null, when in fact the null is true.” It can also be thought of as a “worst-case false-positive rate.” Typically, the researcher would set α in advance before performing the test.

For an ideal test, we want the following to be true:

$$\beta(\theta) \text{ low for } \theta \in \Theta_0, \beta(\theta) \text{ high for } \theta \in \Theta_1, \alpha \text{ low}$$

This would mean that we reject the null often when the alternative is true, and we reject the null infrequently when the null is true. Finally, we have a small maximal probability of rejecting the null when in fact it’s true.

Now, let’s outline the typical framework for frequentist hypothesis testing.

1. State H_0, H_1 for data y .
2. State size α and test-statistic $T(Y)$.
3. Determine critical values to determine critical region.
4. Make decision to reject or retain H_0 based on value of $T(y)$ computed from the data.

8.2 Carrying out hypothesis tests

Let’s begin by examining an example of the power of a test.

Example 8.2.6: Power in a Normal example

Suppose we have an estimator $\hat{\theta}$ such that we have the pivot

$$\frac{\sqrt{n}(\hat{\theta} - \theta)}{\sigma} \sim \mathcal{N}(0, 1).$$

Let σ^2 be known and define the hypotheses

$$H_0 : \theta = 0, H_1 : \theta \neq 0.$$

We will build a test statistic as follows:

$$T(\mathbf{Y}) = \frac{\sqrt{n}\hat{\theta}}{\sigma} = \frac{\sqrt{n}(\hat{\theta} - \theta)}{\sigma} + \frac{\sqrt{n}\theta}{\sigma} \sim \mathcal{N}\left(\frac{\sqrt{n}\theta}{\sigma}, 1\right)$$

Note the distribution of the test-statistic varies depending on which hypothesis is true:

$$T(\mathbf{Y}) \underset{H_0}{\sim} \mathcal{N}(0, 1), T(\mathbf{Y}) \underset{H_1}{\sim} \mathcal{N}\left(\frac{\sqrt{n}\theta}{\sigma}, 1\right)$$

Then, we will perform a two-sided test, where we reject the null if $T(\mathbf{y}) < c_L$ or $T(\mathbf{y}) > c_U$. Thus, we can define the power function of our test as follows:

$$\begin{aligned} \beta(\theta) &= \mathbb{P}(\mathbf{Y} \in A^c) \\ &= \mathbb{P}(T(\mathbf{Y}) > c_U) + \mathbb{P}(T(\mathbf{Y}) < c_L) \\ &= \boxed{1 - \Phi\left(c_U - \frac{\sqrt{n}\theta}{\sigma}\right) + \Phi\left(c_L - \frac{\sqrt{n}\theta}{\sigma}\right)} \end{aligned}$$

Here is figure that visualizes the power function of the example above for varying sample sizes:

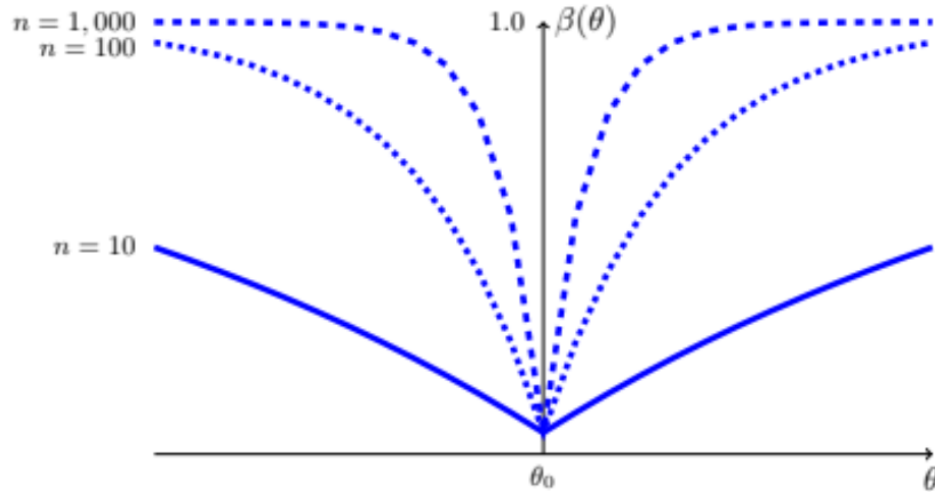


Figure 3: Power function $\beta(\theta)$ for a two-sided test $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ as n increases.

We notice a few interesting things from the figure. First, we see that as the true value of θ strays further from θ_0 , our power increases. This is intuitive, as a good test should reject the null more often as our true θ deviates from the null value. Second, we see that increasing the sample size n makes our power function more sharp. This is also intuitive, as more data means more information that allows us to reject the null more robustly.

One important step we left out in the example above was how to calculate the critical values of our test. We do this below:

Example: Finding critical values for simple null

Suppose we have data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, \sigma^2)$, where σ^2 is known. We define the hypotheses:

$$H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0.$$

Since the Normal is an NEF, we know that the MLE of θ is $\bar{Y}$, so we define the test-statistic

$$T(\mathbf{Y}) = \frac{\sqrt{n}(\bar{Y} - \theta_0)}{\sigma}$$

Then, under H_0 , we know $T(\mathbf{Y}) \sim \mathcal{N}(0, 1)$, a pivot! So, by symmetry of the Normal, we can preset our size α and solve for the corresponding critical values:

$$\alpha = \mathbb{P}(|T| \geq c) = 2\Phi(-c) \Rightarrow \boxed{c = \Phi^{-1}(1 - \alpha/2)}$$

If $\alpha = 0.05$, we would simply use the 0.975 quantile

$$c \approx 1.96$$

We now examine the t-test, a widely-used test to compare the sample mean to some pre-set true mean.

Definition: One sample t-test

We define the t-statistic for data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$.

$$T(\mathbf{Y}) = \frac{\sqrt{n}(\bar{Y} - \mu)}{\hat{\sigma}} \sim t_{n-1},$$

where $\hat{\sigma}^2$ is the sample standard deviation. Since $T(\mathbf{Y})$ is a pivotal quantity, we can carry out a one-sided test of $H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$ with a size of α by rejecting the null if

$$T(\mathbf{Y}) > Q_{t_{n-1}}(1 - \alpha)$$

For a two-sided test of $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ with a size of α by rejecting the null if

$$T(\mathbf{Y}) > Q_{t_{n-1}}(1 - \alpha/2) \text{ or } T(\mathbf{Y}) < Q_{t_{n-1}}(\alpha/2)$$

Note that to perform the test, we have assumed Normal data, which may or may not be a valid assumption.

We can also use asymptotic pivots in order to construct a test.

Example 8.4.4: Approximate result for t-statistic

Suppose we define the hypotheses

$$H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$$

Recall the t-statistic

$$T(\mathbf{Y}) = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})} \xrightarrow[H_0]{d} \mathcal{N}(0, 1)$$

Then, we can find our test size using the limit:

$$\alpha = \lim_{n \rightarrow \infty} \mathbb{P}(|T| \geq c; \theta_0)$$

So, for sufficiently large n we can use the asymptotic critical value c to accept/reject the null.

The Neyman-Pearson lemma gives us an elegant result for any test with both a simple null and a simple alternative.

Lemma: Neyman-Pearson lemma

Suppose we have a simple null and simple alternative, where

$$H_0 : \theta = \theta_0, H_1 : \theta = \theta_1$$

Then, for any size α , the most powerful test uses the likelihood ratio, rejecting H_0 if

$$\frac{f(\mathbf{y}; \theta_1)}{f(\mathbf{y}; \theta_0)} = \frac{\mathcal{L}(\theta_1; \mathbf{Y})}{\mathcal{L}(\theta_0; \mathbf{Y})}$$

is large.

8.3 p-values

We will discuss the definition and properties of p-values in hypothesis testing.

Definition: p-value for simple null

For a simple null hypothesis H_0 , suppose we have a test-statistic $T(\mathbf{Y})$ such that we reject the null when $T(\mathbf{Y})$ is large. Let $t = T(\mathbf{y})$ be the observed value of the test-statistic from the data $\mathbf{y}$. Then, the observed p-value is the statistic

$$p(\mathbf{y}) = \mathbb{P}_{\mathbf{Y}}(T(\mathbf{Y}) \geq t; H_0) = 1 - F_{T(\mathbf{Y})}(t)$$

Intuitively, this is the probability of seeing the observed value of the test-statistic or more extreme under the null. The “direction of extremity” could also be in the opposite direction, if we define a test where we reject H_0 when $T(\mathbf{y})$ is small.

To better understand the properties of p-values, let’s review the Universality of the Uniform.

Example: Universality of the Uniform in hypothesis testing

Suppose we have the simple null $H_0 = \theta = \theta_0$ and a continuous test-statistic T . Assume that $T \sim F^*$. Then by Universality of the Uniform,

$$\boxed{F^*(T) \underset{H_0}{\sim} \text{Unif}(0,1)} \Rightarrow \boxed{1 - F^*(T) \underset{H_0}{\sim} \text{Unif}(0,1)}$$

Next, we show that p-values are distributed Uniform under a simple null.

Theorem 8.7.2: p-values are Uniform under the null

Let $T(\mathbf{Y})$ be a continuous test statistic. We will use a hypothesis test that rejects $H_0 : \theta =$

θ_0 when $T(\mathbf{y})$ is large. Then,

$$p(\mathbf{Y}) \underset{H_0}{\sim} \text{Unif}(0, 1)$$

Proof: Let t_0 be the observed value of T . Then, the associated p-value is $\mathbb{P}(T \geq t_0) = 1 - F_T(t_0)$. So, as a random variable, the p-value is $1 - F_T(T)$. By Universality of the Uniform, the p-value is $\text{Unif}(0, 1)$.

The fact that p-values are distributed $\text{Unif}(0, 1)$ under the null also implies that

$$\mathbb{P}(p(\mathbf{Y}) \leq \alpha) = \alpha.$$

In practice, we set an arbitrary threshold α to reject or retain a null.

8.4 Likelihood-based tests

There are three tests which can be derived from the likelihood function using a simple null and composite alternative:

$$H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$$

Definition: Wald test

The Wald test uses the asymptotic distribution of the MLE to construct a test-statistic. Recall that under some regularity conditions,

$$\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_Y(\theta)^{-1}).$$

Then, for sufficiently large n , we have the approximate pivot

$$\sqrt{\mathcal{I}_Y(\theta)}(\hat{\theta}_{\text{MLE}} - \theta) \sim \mathcal{N}(0, 1).$$

Under the null, our true θ^* is θ_0 , so we can define the test-statistic

$$T(\mathbf{Y}) = \sqrt{\mathcal{I}_Y(\theta)}(\hat{\theta}_{\text{MLE}} - \theta_0) \sim \mathcal{N}(0, 1)$$

Then, for size α , we reject H_0 if

$$T(\mathbf{Y}) < \Phi^{-1}(\alpha/2) \text{ or } \Phi^{-1}(1 - \alpha/2)$$

Definition: Score test

The score test uses the score function evaluated at the null value. Recall the score function

$$s(\theta; \mathbf{Y}) = \frac{\partial}{\partial \theta} \log \mathcal{L}(\theta; \mathbf{Y})$$

Under the null, $\theta^* = \theta_0$, meaning

$$\mathbb{E}[s(\theta_0; \mathbf{Y})] = 0$$

Recall also that by the Information equality,

$$\text{Var}(s(\theta_0; \mathbf{Y})) = \mathcal{I}_{\mathbf{Y}}(\theta_0)$$

Then, recall that score is asymptotically Normal, meaning

$$s(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, \mathcal{I}_{\mathbf{Y}}(\theta_0))$$

Then, we define the following approximate pivot as our test-statistic:

$$T(\mathbf{Y}) = \frac{s(\theta_0; \mathbf{Y})}{\sqrt{\mathcal{I}_{\mathbf{Y}}(\theta_0)}} \sim \mathcal{N}(0, 1)$$

Thus, for size α , we reject H_0 if

$$T(\mathbf{Y}) < \Phi^{-1}(\alpha/2) \text{ or } \Phi^{-1}(1 - \alpha/2)$$

The score test is convenient in the fact that we don't need to compute $\hat{\theta}_{\text{MLE}}$, which could be difficult in practice.

Definition: Likelihood ratio test

The likelihood ratio test uses the likelihood ratio, defined below

$$\frac{\mathcal{L}(\hat{\theta}_{\text{MLE}}; \mathbf{Y})}{\mathcal{L}(\theta_0; \mathbf{Y})} \geq 1$$

In practice, the log-likelihood ratio is easier to use

$$\log \mathcal{L}(\hat{\theta}_{\text{MLE}}; \mathbf{Y}) - \log \mathcal{L}(\theta_0; \mathbf{Y})$$

Then define the following test-statistic

$$\Lambda(\mathbf{Y}) = 2 [\log \mathcal{L}(\hat{\theta}; \mathbf{Y}) - \log \mathcal{L}(\theta_0; \mathbf{Y})] \xrightarrow{d} \chi_1^2 \text{ by Thm 8.6.2}$$

Then, we reject the null with size α if

$$\Lambda(\mathbf{Y}) > Q_{\chi_1^2}(1 - \alpha)$$

We compare the three test in the figure below:

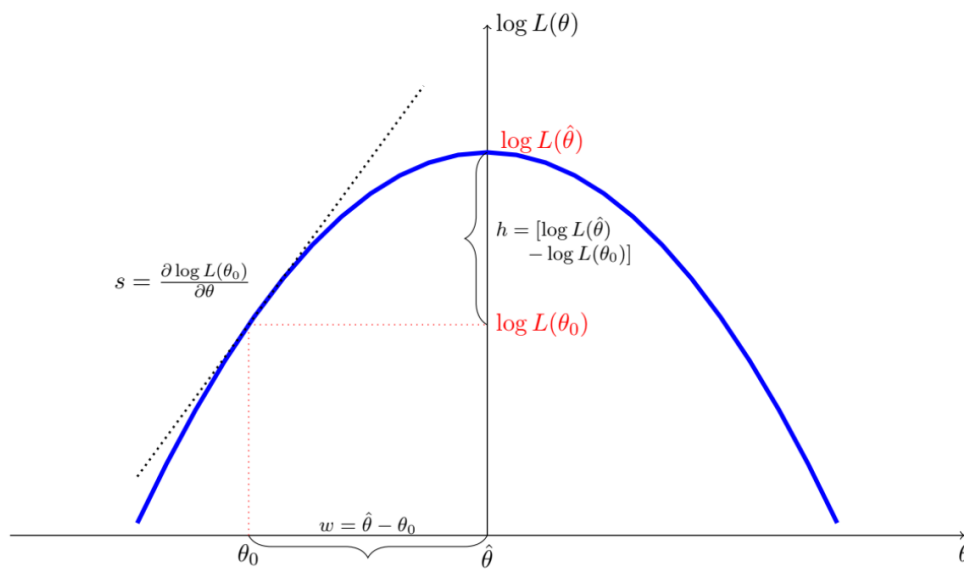


Figure 4: Comparison of the Wald (based on distance $w = \hat{\theta} - \theta_0$), score (based on slope $s = s(\theta_0) = \partial \log \mathcal{L}(\theta_0) / \partial \theta$), and LLR test (based on distance $w = \hat{\theta} - \theta_0$).

9 Bayesian statistics

9.1 Bayesian thinking

In the frequentist framework, we treated θ as a fixed value, not a random variable. This meant that we had to a lot of bending-over-backwards to avoid making probability statements about θ . For example, when we defined a CI for θ , the interval itself was random! Under the Bayesian framework we will treat θ as an r.v., then write down a "full probability model."

We'll work with a parametric model $F_{Y|\theta}$ for data $\mathbf{Y}$. Note the conditioning bar, which we can use instead of $;$ since θ is now an r.v. Bayesian thinking allows us to default back to probability theory, following the two principles below:

1. Always obey probability laws
2. Model all uncertainty using probability

Bayesian statistics is based on Bayes' rule. The idea is to use Bayes' rule to obtain the posterior distribution of θ , the belief we have in θ after having seen the data $\mathbf{y}$.

Definition: Posterior distribution

We apply Bayes' rule to obtain the posterior distribution of θ , denoted $\pi(\theta|\mathbf{y})$.

$$\pi(\theta|\mathbf{y}) = \frac{f(\mathbf{y}|\theta)\pi(\theta)}{f(\mathbf{y})} \propto \mathcal{L}(\theta;\mathbf{y})\pi(\theta)$$

Note that since the marginal likelihood $f(\mathbf{y})$ is a normalizing constant, we can write the posterior as proportional to the likelihood times the prior, which quantifies our belief in θ prior to seeing the data. The marginal likelihood can always be calculated:

$$f(\mathbf{y}) = \int_{-\infty}^{\infty} f(\mathbf{y}|\theta)\pi(\theta)d\theta$$

What's nice about the Bayesian approach is that there's no controversy to our calculation above. Once we've defined a prior $\pi(\theta)$ and a model $f_{Y|\theta}(\mathbf{y})$, we simply apply the laws of probability. The caveat is that writing down a prior and a model is a high barrier of entry, and is difficult in practice.

Let's take a look at an example of how Bayes' rule can be applied to obtain the posterior distribution $\theta|\mathbf{y}$.

Example 9.2.3 in textbook: Single Bernoulli trial with Beta prior

Assume that our data is Y_1 . Our model is $Y_1|\theta \sim \text{Bern}(\theta)$, and our prior is $\theta \sim \text{Beta}(\alpha, \beta)$. Let's write down the likelihood function:

$$\mathcal{L}(\theta; Y_1) = \mathbb{P}(Y_1 = y_1|\theta) = \theta^{y_1}(1 - \theta)^{1-y_1}$$

Thus, applying the Beta PDF and Bayes' rule, we can write down the posterior distri-

bution $\pi(\theta|\mathbf{y})$.

$$\begin{aligned}\pi(\theta|\mathbf{y}) &\propto \mathcal{L}(\theta; Y_1)\pi(\theta) \\ &\propto \left\{ \theta^{y_1}(1-\theta)^{1-y_1} \right\} \left\{ \theta^{\alpha-1}(1-\theta)^{\beta-1} \right\} \text{ by Beta PDF} \\ &= \theta^{y_1+\alpha-1}(1-\theta)^{1-y_1+\beta-1}\end{aligned}$$

This is simply the $\text{Beta}(\alpha + y_1, \beta + 1 - y_1)$ PDF, without the normalizing constant! Thus, we have the following result.

$$\boxed{\theta|\mathbf{y} \sim \text{Beta}(\alpha + y_1, \beta + 1 - y_1)}$$

Note that this result is an example of Beta-Binomial conjugacy! When a Beta prior is fed through a Binomial model, we still get a Beta posterior.

We can actually generalize the result above to n Bernoulli datapoints:

Example 9.2.3 in textbook (cont.): n ind. Bernoulli trials with Beta prior

Assume our data is $\mathbf{Y} = Y_1, \dots, Y_n$, where $Y_j|\theta \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(\theta)$, with prior $\theta \sim \text{Beta}(\alpha, \beta)$. We have the likelihood function below:

$$\begin{aligned}\mathcal{L}(\theta, \mathbf{Y}) &= \prod_{j=1}^n f_{Y_j|\theta}(y_j) \text{ by ind.} \\ &= \theta^{n\bar{y}}(1-\theta)^{n(1-\bar{y})} \text{ by Bern. PMF}\end{aligned}$$

Plugging in this likelihood into the expression for $\pi(\theta|\mathbf{y})$, we obtain the result

$$\pi(\theta|\mathbf{y}) = \theta^{\alpha+n\bar{y}-1}(1-\theta)^{\beta+n(1-\bar{y})-1}$$

Thus, by the Beta PDF,

$$\boxed{\theta|\mathbf{y} \sim \text{Beta}(\alpha + n\bar{y}, \beta + n(1 - \bar{y}))}$$

Priors must be constructed carefully. Note the following:

Remark: Cromwell's rule

"When constructing a prior, don't set $\pi(\theta)$ for any value of θ !" This is because if we let the prior be 0 anywhere, the posterior $\theta|\mathbf{y}$ will always be 0 no matter how much the data might suggest that that value of θ is a good fit for the data.

9.2 Bayesian point estimators

The “output” of the Bayesian workflow is the posterior distribution $\pi(\theta|\mathbf{y})$. But, it’s often easier to work with point values rather than the entire posterior distribution. In other words, we want a clear summary of our output. Some common Bayesian point estimators are listed below:

Definition: Posterior mean, median, mode

Suppose we’ve obtained the posterior distribution $\pi(\theta|\mathbf{y})$. We define the following quantities.

- 1) Posterior mean: $\mathbb{E}[\theta|\mathbf{y}] = \int_{-\infty}^{\infty} \theta \pi(\theta|\mathbf{y}) d\theta$
- 2) Posterior median: $Q_{\theta|\mathbf{y}}(0.5)$
- 3) Posterior mode: $\operatorname{argmax}_{\theta \in \Theta} \pi(\theta|\mathbf{y})$

The posterior mode is also called the maximum a posteriori (MAP) estimator.

Often, we’ll define a Bayesian estimator $\hat{\theta}$ as a quantity that minimizes the posterior expected loss function, a function we defined to measure the distance between our true θ and our estimate $\hat{\theta}$.

Definition: Posterior expected loss

We construct a general Bayesian estimator $\hat{\theta}_{\text{Bayes}}$ using a loss function $\text{Loss}(\theta, \hat{\theta})$.

$$\hat{\theta}_{\text{Bayes}} = \operatorname{argmin}_{\hat{\theta} \in \Theta} \mathbb{E}[\text{Loss}(\theta, \hat{\theta})|\mathbf{y}]$$

Our estimator minimizes the posterior expected loss: the expected loss given the data $\mathbf{y}$, which can be computed using the posterior distribution $\theta|\mathbf{y}$.

We can actually derive the posterior mean and mode using the definition above.

Theorem 9.3.3 in textbook: Origins of posterior mean, median

If we use the squared-error loss, $\text{Loss}(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$, then the posterior mean minimizes the posterior expected loss:

$$\mathbb{E}[\theta|\mathbf{y}] = \operatorname{argmin}_{c \in \Theta} \mathbb{E}[(\theta - c)^2|\mathbf{y}]$$

If we used the absolute error loss, $\text{Loss}(\theta, \hat{\theta}) = |\theta - \hat{\theta}|$, then the posterior median mini-

mizes the posterior expected loss:

$$Q_{\theta|y}(0.5) = \operatorname{argmin}_{c \in \Theta} \mathbb{E}[|\theta - c| | y]$$

Using some clever tricks, we can also derive a statement about the posterior mean for Gaussian data that generalizes to all possible priors.

Example 9.3.4 in textbook: Tweedie's formula

Assume Gaussian data Y , such that

$$f_{Y|\theta}(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y - \theta)^2 \right\}$$

Then for any prior on θ , the following statement holds:

$$\mathbb{E}[\theta|y] = y + \sigma^2 \frac{\partial \log f(y)}{\partial y}$$

Tweedie's formula is fascinating, because it means that for Gaussian data, we don't need any information about θ to obtain the posterior mean!

Next, let's explore some properties of the MAP estimator. This estimator is convenient because we don't ever need to compute the normalizing constant. This is because we don't need the posterior distribution $\theta|y$. We can simply compute the MAP estimate using the likelihood and prior.

Remark: Properties of the MAP estimator

Recall the definition of the MAP estimator:

$$\hat{\theta}_{\text{MAP}} = \operatorname{argmax}_{\theta \in \Theta} \pi(\theta|y)$$

We can express the posterior distribution in terms of the likelihood, prior, and some constant c .

$$\pi(\theta|y) \propto \mathcal{L}(\theta; y) \pi(\theta) \Rightarrow \log \pi(\theta|y) = c + \log \mathcal{L}(\theta; y) + \log \pi(\theta)$$

Thus, we can rewrite the MAP in terms of only the likelihood and prior:

$$\begin{aligned} \hat{\theta}_{\text{MAP}} &= \operatorname{argmax}_{\theta \in \Theta} \log \pi(\theta|y) \\ &= \operatorname{argmax}_{\theta \in \Theta} \{c + \log \mathcal{L}(\theta; y) + \log \pi(\theta)\} \\ &= \operatorname{argmax}_{\theta \in \Theta} \{\log \mathcal{L}(\theta; y) + \log \pi(\theta)\} \end{aligned}$$

Note that if our prior is Uniform (constant with respect to θ), then the MAP is simply the MLE!

9.3 Credible intervals

In frequentist statistics, we had the confidence interval. For Bayesian statistics, we have the credible interval.

Definition: Credible interval

We define a $1 - \alpha$ credible interval for θ using the constants $L(\mathbf{y})$ and $U(\mathbf{y})$.

$$\mathbb{P} \{L(\mathbf{y}) \leq \theta \leq U(\mathbf{y}) | \mathbf{y}\} = 1 - \alpha$$

The credible interval is exactly the opposite of a confidence interval. In a confidence interval, θ was fixed and our bounds $L(\mathbf{Y})$ and $U(\mathbf{Y})$ were r.v.s. In our credible interval, θ is the r.v. and $L(\mathbf{y})$ and $U(\mathbf{y})$ are constants. In the Bayesian lens, we can make a direct probability statement about where θ is located!

We can define a credible interval using the quantile function as we previously did for a confidence interval.

Remark: Defining credible interval with quantiles

In general, for a $1 - \alpha$ credible interval, we can define the interval

$$[Q_{\theta|\mathbf{y}}(\alpha/2), Q_{\theta|\mathbf{y}}(1 - \alpha/2)]$$

This is equivalent to us taking a central "slice" of the posterior distribution $\pi(\theta|\mathbf{y})$, for which the area under the curve is $1 - \alpha$.

9.4 Computing Bayesian estimators via simulation

Although Bayesian estimators have nice probabilistic properties and are straightforward to analyze in theory, they are difficult to actually simulate. They often involve hard integrals like the following:

Example: Computing the posterior mean

Calculating Bayesian estimators often leads to computational challenges, such as very hard integrals!

$$\mathbb{E}[\theta|\mathbf{y}] = \int \theta \pi(\theta|\mathbf{y}) d\theta = \frac{\int \theta \pi(\theta) \mathcal{L}(\theta; \mathbf{y}) d\theta}{\int \pi(\theta) \mathcal{L}(\theta; \mathbf{y}) d\theta}$$

We need a better way to do this. Simulation seems promising. We know that Markov-chain Monte Carlo (MCMC) can simulate draws from a given continuous distribution f . We can apply this to our posterior distribution to obtain estimates of our Bayesian estimators.

Example: Bayesian simulation

Say we have the posterior distribution $\pi(\theta|\mathbf{y})$. We want to compute 2 Bayesian estimators: the posterior mean $\mathbb{E}[\theta|\mathbf{y}]$ and the posterior median $Q_{\theta|\mathbf{y}}(0.5)$. However, these integrals are too hard! We simply sample draws from $\pi(\theta|\mathbf{y})$ using MCMC:

$$\theta_1^*, \dots, \theta_B^* \sim \theta|\mathbf{y}$$

Then we estimate the posterior mean and posterior median from the data!

$$\mathbb{E}[\theta|\mathbf{y}] \approx \frac{1}{B} \sum_{b=1}^B \theta_b^*, \quad Q_{\theta|\mathbf{y}}(0.5) \approx \hat{Q}_{\lceil 0.5B \rceil}(0.5)$$

Note that this strategy is invariant under reparameterization of θ .

Example: Bayesian simulation using reparameterization

Say our friend wants to calculate the mean posterior mean and median of λ instead of θ , where $\lambda = h(\theta)$. Then, we simply apply the transformation h to our simulated draws from $\pi(\theta|\mathbf{y})$.

$$\lambda_b^* = h(\theta_b^*) \text{ for } b = 1, \dots, B$$

Then, we again just estimate the posterior mean and posterior median from the data.

$$\mathbb{E}[\lambda|\mathbf{y}] \approx \frac{1}{B} \sum_{b=1}^B h(\theta_b^*), \quad Q_{\lambda|\mathbf{y}}(0.5) \approx \hat{Q}_{\lceil 0.5B \rceil}(0.5)$$

We can do all of this without worrying about change of variables or the Jacobian, which is very convenient.

Simulation is especially powerful when our parameter θ is multivariate: $\theta = (\psi, \lambda)$. Suppose we want to study λ alone.

Example: Bayesian simulation using marginalization

Suppose $\theta = (\psi, \lambda)$, and we've simulated draws from $\pi(\theta|\mathbf{y})$:

$$\theta_1^*, \dots, \theta_B^* = \begin{pmatrix} \psi_1^* \\ \lambda_1^* \end{pmatrix}, \dots, \begin{pmatrix} \psi_B^* \\ \lambda_B^* \end{pmatrix} \sim \theta|\mathbf{y}$$

We would like to study λ , so we want $\pi(\lambda|\mathbf{y}) = \int \pi(\psi, \lambda|\mathbf{y}) d\psi$ by marginalization. This is very hard to compute. So, instead, we marginalize by just throwing out the ψ_b^* . We're left with

$$\lambda_1^*, \dots, \lambda_B^* \sim \lambda|\mathbf{y}$$

Now, we can estimate quantities such as $\mathbb{E}[\lambda|\mathbf{y}]$ or $Q_{\lambda|\mathbf{y}}(0.5)$ just as we did above.

9.5 Conjugate priors

It turns out that some priors and likelihoods work especially well together if they have a particular distribution. We've seen conjugacy in Stat 110 already, so let's explore it deeper.

Definition: Conjugate prior

Recall Bayes' rule:

$$\pi(\theta|\mathbf{y}) \propto \pi(\theta)\mathcal{L}(\theta;\mathbf{y})$$

A family of priors $\pi(\theta)$ is conjugate for a statistical model if choosing a prior in that family leads to a posterior also in the same family.

We've seen a couple examples of this in the past.[ADD THE N DATAPOINTS VERSION]

Example: Beta-Binomial and Gamma-Poisson conjugacy

Beta-Binomial conjugacy:

$$Y|p \sim \text{Bin}(n, p), p \sim \text{Beta}(a, b) \Rightarrow p|(Y = y) \sim \text{Beta}(a + y, b + n - y)$$

Gamma-Poisson conjugacy:

$$Y|\lambda \sim \text{Pois}(\lambda), \lambda \sim \text{Gamma}(r_0, b_0) \Rightarrow \lambda|(Y = y) \sim \text{Gamma}(r_0 + y, b_0 + 1)$$

Next, we extend this to examine conjugacy involving the Normal.

Example 9.5.4 in textbook: Normal-Normal conjugacy (single observation)

Suppose we have a single observation Y . We define $Y|\mu \sim \mathcal{N}(\mu, \sigma^2)$ and $\mu \sim \mathcal{N}(m_0, \tau_0^2)$. Assume σ^2, m_0, τ_0^2 known. We want to find the posterior $\mu|y$. By Bayes' rule,

$$\pi(\mu|y) = c \cdot \pi(\mu)\mathcal{L}(\mu; y) \text{ where } c \text{ is a constant}$$

We examine the log posterior,

$$\begin{aligned} \log \pi(\mu|y) &= c + \log \pi(\mu) + \log \mathcal{L}(\mu; y) \\ &= c' - \frac{1}{2\tau_0^2}(\mu - m_0)^2 + \frac{1}{2\sigma^2}(y - \mu)^2 \\ &= c'' - \frac{\mu^2}{2} \left(\frac{1}{\sigma^2} - \frac{1}{\tau_0^2} \right) + \mu \left(\frac{y}{\sigma^2} + \frac{m_0}{\tau_0^2} \right) \\ &= c'' - \frac{\mu^2}{2\tau_1^2} + \frac{\mu}{\sigma_1^2} m_1 \end{aligned}$$

Here, we've made the following substitutions:

$$\tau_1^{-2} = \sigma^{-2} + \tau_0^{-2}, \quad m_1 = \tau_1^2 \left(\frac{y}{\sigma^2} + \frac{m_0}{\tau_0^2} \right)$$

Thus, pattern-matching to the log density of the Normal,

$$\boxed{\mu|y \sim \mathcal{N}(m_1, \tau_1^2)}$$

Note that the σ^{-2} and τ_0^2 are the precisions, meaning that the precisions of the likelihood and prior add!

We can utilize the same logic to examine Normal-Normal conjugacy for n datapoints.

Theorem 9.5.5 in textbook: Normal-normal conjugacy (n observations)

We generalize the above to n observations. Suppose we have data $Y_1, \dots, Y_n$, where $Y_j|\mu \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$. Then,

$$\boxed{\mu|\mathbf{y} \sim \mathcal{N}(\mu_n, \tau_n^2)},$$

where

$$\tau_n^{-2} = n\sigma^{-2} + \tau_0^{-2}, \quad \mu_n = \tau_n^2 \left(\frac{n\bar{y}}{\sigma^2} + \frac{\mu_0}{\tau_0^2} \right)$$

Note that if we define a new quantity b_n as the quotient of the posterior variance and the prior variance,

$$b_n = \frac{\tau_n^2}{\tau_0^2} = \frac{\tau_0^{-2}}{n\sigma^{-2} + \tau_0^{-2}} = \frac{\sigma^2}{\sigma^2 + n\tau_0^2}$$

then we can rewrite the posterior distribution nicely as

$$\mu|\mathbf{y} \sim \mathcal{N}((1 - b_n)\bar{y} + b_n\mu_0, b_n\tau_0^2)$$

Now, note that the posterior mean is a weighted average of the prior mean μ_0 and the sample mean $\bar{y}$:

$$\mathbb{E}[\mu|\mathbf{y}] = (1 - b_n)\bar{y} + b_n\mu_0$$

Here, we also call b_n the shrinkage factor. We see that more data and higher n means b_n will decrease, which puts more weight on the sample mean compared to the prior mean. This matches our intuition, as the more data we collect, the less the prior matters.

9.6 Predictive posterior

Bayesian thinking can also be applied to out of sample prediction. In the frequentist world, we usually have data $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} f(y; \theta)$. Say we want to predict some Y_{new} given the data we already have. An intuitive r.v. might be $Y_{\text{new}}|Y_1, \dots, Y_n$, but this means nothing because we assumed i.i.d.! Maybe we tried to sample from $f(y_{\text{new}}; \hat{\theta})$ instead to generate our next Y_{new} . But now, we're completely ignoring our uncertainty about θ , compressing it into a single value that we've estimated.

The Bayesian approach solves our problem! Since we're able to write down a full probability model for our current data $\mathbf{y}$ in addition to our parameter θ , we can apply these to think about the distribution of $Y_{\text{new}}|\mathbf{y}$.

Definition: Posterior predictive distribution

Suppose we've observed data $\mathbf{y}$, and we wish to use this data to predict new data Y_{new} . Then, the posterior predictive distribution of Y_{new} is

$$Y_{\text{new}}|\mathbf{y}$$

The posterior predictive distribution can be derived using LOTP.

$$f(Y_{\text{new}}|\mathbf{y}) = \int f(Y_{\text{new}}|\mathbf{y}, \theta) \pi(\theta|\mathbf{y}) d\theta$$

As we see above, the posterior predictive distribution can (sometimes) be derived analytically. The $f(Y_{\text{new}}|\mathbf{y}, \theta)$ term depends on the parametric model we specify, and often times it ends up being equal to $f(Y_{\text{new}}|\theta)$ if we specify that $Y_j|\theta$ are conditional independent. The $\pi(\theta|\mathbf{y})$ term is simply our posterior that we would have derived by this point.

An analytical solution might not always be feasible. So, we turn to simulation!

Example: Posterior predictive distribution via simulation

Suppose we've simulated the following from the posterior $\pi(\theta|\mathbf{y})$:

$$\theta_1^*, \dots, \theta_B^* \sim \theta|\mathbf{y}$$

Then we can simply generate a draw from $f(y|\theta_b^*)$ for each $b = 1, \dots, B$.

$$y_{\text{new},1}^* \sim y|\theta_1^*, \dots, y_{\text{new},B}^* \sim y|\theta_B^*$$

This is sort of a nested simulation idea! By doing this we're able to "propagate" our uncertainty about θ all the way through to the sampled y_{new} .

Let's do an example, where we'll apply Gamma-Poisson conjugacy in order to find the posterior predictive distribution $Y_{\text{new}}|\mathbf{y}$.

Example: Posterior predictive distribution and Gamma-Poisson conjugacy

Suppose we have a Poisson process with rate λ . We impose a prior $\lambda \sim \text{Gamma}(r_0, b_0)$. We see Y arrivals within the fixed time interval t . Note that $Y|\lambda \sim \text{Pois}(\lambda t)$. Find the posterior predictive distribution $Y_{\text{new}}|y$.

We begin by solving for the posterior distribution. By Gamma-Poisson conjugacy,

$$\lambda|y \sim \text{Gamma}(r_0 + y, b_0 + t)$$

Now, let's solve for the marginal density of Y . By Story 8.4.5 in the Stat 110 textbook,

$$Y \sim \text{NBin}\left(r_0, \frac{b_0}{b_0 + t}\right)$$

Let's summarize what we've done so far to obtain the marginal density of Y . A Gamma prior on λ and a Poisson model for the data given λ yielded a Negative Binomial marginal for the data. Now, the idea is to reiterate the process, except we now condition everything on having already incorporated the y . This will allow us to find the density of $Y_{\text{new}}|y$, since we're finding the "marginal" of new data within the world where we've already seen y .

Now, we use our updated $\lambda|y \sim \text{Gamma}(r_0 + y, b_0 + t)$ as the new prior, and our model for the data stays the same as $Y_{\text{new}}|\lambda, y \sim \text{Pois}(\lambda t_{\text{new}})$. Thus, we apply the result from Story 8.4.5 once again to obtain the posterior predictive distribution.

$$Y_{\text{new}}|y \sim \text{NBin}\left(r_0 + y, \frac{b_0 + t}{b_0 + t + t_{\text{new}}}\right)$$

The fact of the marginal density of the data being Negative-Binomial for a Poisson model with Gamma prior is incredibly useful.*

9.7 Hierarchical models

Hierarchical modeling allows us to describe an n -step data generating process that allows us to model each "step" of the process separately.

Definition: Two-level Gaussian hierarchical model

Suppose we model our data $Y_1, \dots, Y_K$, defining $Y_j|\mu \stackrel{\text{ind.}}{\sim} \mathcal{N}(\mu_j, \sigma^2)$, where $\mu = (\mu_1, \dots, \mu_K)$. Then, we also set a distribution for each condition mean $\mu_j \stackrel{\text{ind.}}{\sim} \mathcal{N}(\gamma, \lambda_0^2)$. Then, the pair (Y, μ) follow a two-level hierarchical model. Here, we call $(\sigma, \gamma, \lambda_0)$ hyperparameteres, treating them as known.

We can think of the above as generating a μ , then using that μ to generate Y .

[DEVELOP MORE INTUITION]

9.8 Decision theory

We'll return to examine frequentists methods for evaluating estimators. We previously saw that finding some estimators can be formulated as a loss minimization problem. Here, we define a related quantity, the risk function:

Definition: Risk function

Say we've defined a loss function $\text{Loss}(\theta, \hat{\theta})$ for evaluating an estimator $\hat{\theta}$. Then the risk function is

$$\text{Risk}_{\hat{\theta}}(\theta) = \mathbb{E}_{\mathbf{Y}}[\text{Loss}(\theta, \hat{\theta}); \theta]$$

In decision theory, we're often interested in finding an optimal estimator based on its risk: the expected loss. We can plot our risk function, which is a function of θ , for several different estimators. Here is an example:

We see that in some regimes of θ , estimator 1 might have the lowest risk, while in other regimes, estimator 2 or 3 might have the lowest risk. So which estimator is best? Sometimes, there's an obvious choice.

Definition: Admissibility/Inadmissibility

An estimator $\hat{\theta}$ is inadmissible if there exists another estimator $\tilde{\theta}$ such that

$$\text{Risk}_{\tilde{\theta}}(\theta) \leq \text{Risk}_{\hat{\theta}}(\theta) \quad \forall \theta \in \Theta$$

Where the inequality is strict for at least one θ . In other words, the estimator $\hat{\theta}$ is inadmissible if there's another estimator that has lower expected loss than it over all θ .

There is a fascinating result in regards to the inadmissibility of an estimator that seems extremely natural.

Theorem: Stein's paradox

Suppose we have data $Y_1, \dots, Y_K \stackrel{\text{ind.}}{\sim} \mathcal{N}(\mu_j, \sigma^2)$, where σ^2 is known and $K \geq 3$. We would like to estimate $\mu = (\mu_1, \dots, \mu_K)$. We use the squared error loss:

$$\text{Loss}(\mu, \hat{\mu}) = \sum_{j=1}^K (\mu_j - \hat{\mu}_j)^2$$

A natural estimator is the MLE of μ , which is

$$\hat{\mu}_{\text{MLE}} = \mathbf{Y} = (Y_1, \dots, Y_K)$$

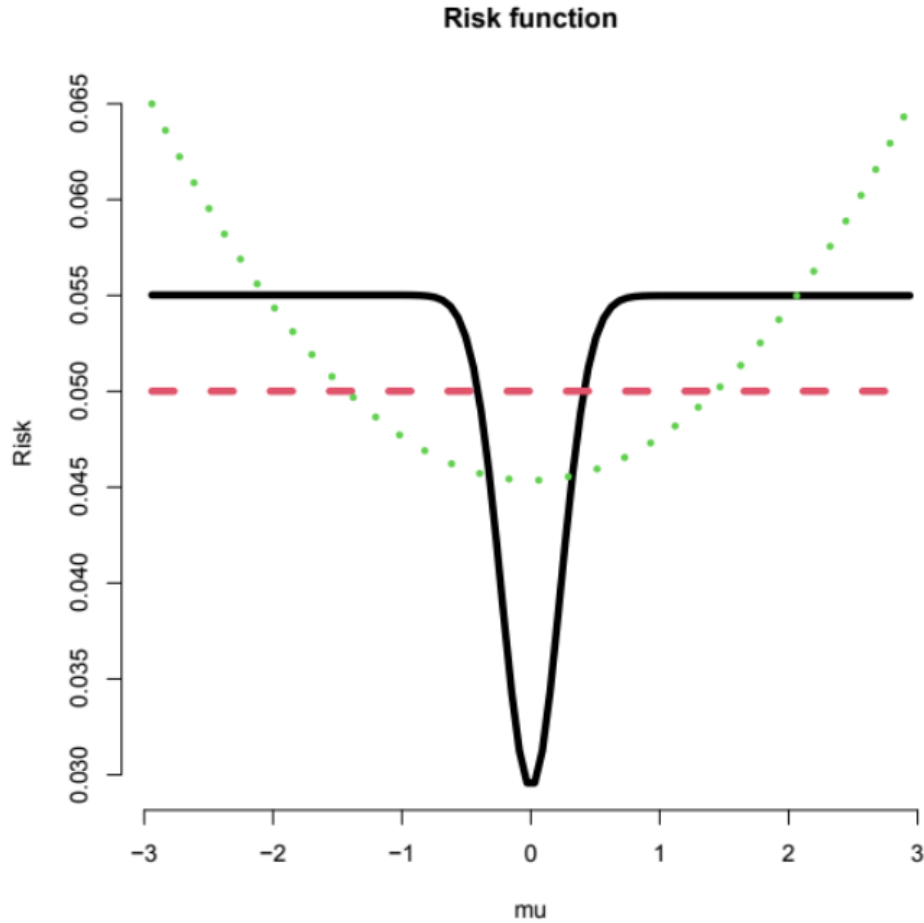


Figure 5: Risk function for 3 different estimators $\hat{\theta}$ of θ .

This can be derived using properties of NEFs, or MoM, but we know it is unbiased. However, Stein's paradox tells us:

The MLE $\mathbf{Y}$ is inadmissible.

Stein also derived an estimator that has strictly lower risk than the MLE for all μ .

Definition: James-Stein Estimator

For the scenario given above, the James-Stein estimator given below has strictly lower risk than the MLE for all μ .

$$\hat{\mu}_{JS,j} = \left(1 - \frac{(K-2)\sigma^2}{\sum_{i=1}^K Y_i^2} \right) Y_j$$

10 Sampling and resampling

10.1 Design-based inference

In the design-based framework, we're interested in random sampling from a fixed, finite population of size N in order to estimate population-level quantities. Our data are the population values y_i for each individual with ID i :

$$y_1, \dots, y_N$$

Here, the $y_1, \dots, y_N$ do not represent crystallizations of an underlying generative model. Instead, the data is simply fixed: just "numbers on a sheet." The randomness will come from the sampling, not the data itself. If we're able to see all data populations values $y_1, \dots, y_N$, then we have a census. Our goal is to estimate population-level quantities:

Definition: Design-based estimands

A design-based estimand is a function of the population value $y_1, \dots, y_N$. Examples are

$$\begin{aligned} 1) \mu &= \frac{1}{N} \sum_{i=1}^N y_i \text{ (population mean)} \\ 2) \sigma^2 &= \frac{1}{N} \sum_{i=1}^N (y_i - \mu)^2 \text{ (population variance)} \end{aligned}$$

10.2 Sampling designs

In the real world, we will have to sample some subset of $y_1, \dots, y_N$ in order to make estimate the design-based estimands above. To do so, we need to construct a random sample.

Definition: Random sample

A random sample of size n can be defined

$$\mathbf{Y} = (Y_1, \dots, Y_n)$$

where Y_j is the value for the j th sampled individual. The Y_j are random, and this randomness stems from the sampling procedure, not any underlying model. The random sample can also be written in terms of $I_j \in \{1, \dots, N\}$, the ID number of the j th individual selected for our sample.

$$\mathbf{Y} = (Y_1, \dots, Y_n) = (y_{I_1}, \dots, y_{I_n})$$

Note here that there is NO connection between Y_i and y_i .

Here is a nice visual for what a random sample does:

The procedure by which we construct a random sample $\mathbf{Y}$ can be defined as below:

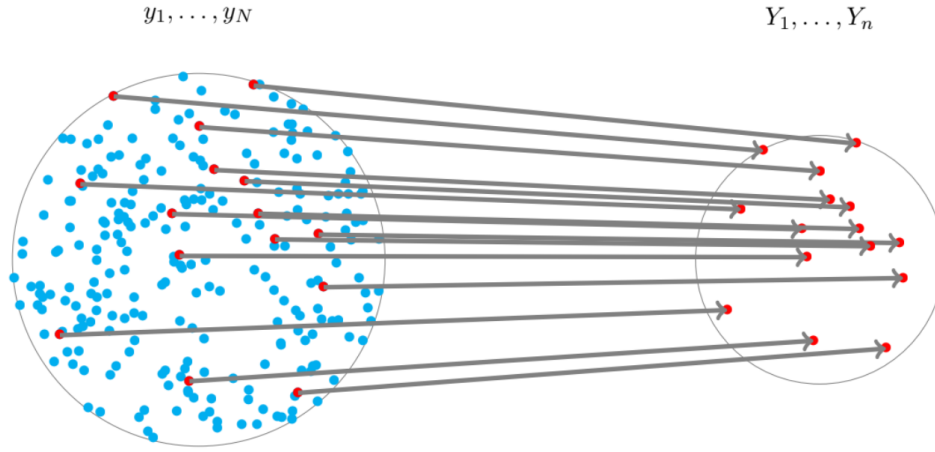


Figure 6: Random sample $\mathbf{Y} = (Y_1, \dots, Y_n)$ from data $y_1, \dots, y_N$.

Definition: Sampling design

Suppose we will collect a random sample of size n , which we write as $\mathbf{Y} = (Y_{I_1}, \dots, Y_{I_n})$. Here $I_j \in \{1, \dots, N\}$ represents the ID number of the j th sampled individual. The sampling design is simply the joint PMF of the I_j :

$$\mathbb{P}(I_1 = i_1, \dots, I_n = i_n)$$

Now, let's examine a few common sampling designs and their properties. The three main designs we will analyze are:

1. Simple random sampling with replacement
2. Simple random sampling without replacement
3. Stratified random sampling

Definition: Simple random sampling (SRS) with replacement

In an SRS with replacement, each of the Y_j has an example probability of being any of the values $y_1, \dots, y_N$. More formally,

$$I_1, \dots, I_n \stackrel{\text{i.i.d.}}{\sim} \text{DUnif}(\{1, \dots, N\}), \text{ where } Y_j = y_{I_j}$$

As a consequence, the Y_j are i.i.d. Intuitively, this means we take a ball out of a jar, then put it back in before sampling the next ball.

Now, an example for SRS with replacement.

Example: Estimating population mean using SRS with replacement

Suppose we'd like to estimate the population mean μ , where

$$\mu = \frac{1}{N} \sum_{i=1}^N y_i$$

We've collected a simple random sample $\mathbf{Y} = (Y_1, \dots, Y_n)$. Suppose we estimate μ using $\bar{Y}$. Let's find the bias and variance of $\hat{\mu} = \bar{Y}$. We begin with the bias:

$$\begin{aligned} \mathbb{E}[\bar{Y}] &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i] \text{ by linearity} \\ &= \frac{1}{n} \sum_{i=1}^n \mu \text{ by symmetry} \\ &= \mu \Rightarrow \boxed{\hat{\mu} \text{ is unbiased!}} \end{aligned}$$

Next, the variance:

$$\begin{aligned} \text{Var}(\bar{Y}) &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i) \text{ by independence} \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 \text{ by symmetry} \\ &= \boxed{\sigma^2/n} \end{aligned}$$

On another note, n can be as big as we want! This is because we're sampling with replacement, meaning we can sample forever. So, we can apply CLT to $\hat{\mu} = \bar{Y}$ as well!

$$\sqrt{n}(\bar{Y} - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Next, we move on to random sampling without replacement.

Definition: SRS without replacement

In an SRS without replacement, we sample sequentially, meaning that Y_j is drawn uniformly from $y_1, \dots, y_N$, Y_2 is drawn from the remaining y_j , etc. Intuitively, we're drawing balls from a jar without returning any to the jar. This implies that Y_j are dependent!

Note that intuitively the Y_j are negatively correlated. This is because the values of $y_1, \dots, y_N$ are already set, meaning that if we choose a large value out of these for Y_1 , then $Y_{2:n}$ will be drawn from a pool that has smaller values on average than the full set of population values.

Now an example.

Example: Estimating population mean using SRS without replacement

Suppose we've collected a sample $\mathbf{Y} = (Y_1, \dots, Y_n)$ and we'd like to estimate the population mean μ using our sample mean $\bar{Y}$. Find the bias and variance of $\hat{\mu} = \bar{Y}$. First the bias:

$$\mathbb{E}[\bar{Y}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i] = \mu \text{ by linearity, symmetry} \Rightarrow \boxed{\hat{\mu} \text{ unbiased}}$$

The variance is tricky because the Y_j are dependent when we sample without replacement. So, we need to expand the variance using covariances:

$$\begin{aligned} \text{Var}(\bar{Y}) &= \frac{1}{n^2} \left\{ n\text{Var}(Y_1) + 2\binom{n}{2}\text{Cov}(Y_1, Y_2) \right\} \text{ by symmetry} \\ &= \frac{1}{n^2} \left\{ n\sigma^2 + 2\binom{n}{2}\text{Cov}(Y_1, Y_2) \right\} \end{aligned}$$

Now, we need to solve for the covariance term $\text{Cov}(Y_1, Y_2)$, which we know depends on $\mathbb{E}[Y_1, Y_2]$. Let's solve for this first, then plug back in.

$$\begin{aligned} \mathbb{E}[Y_1 Y_2] &= \mathbb{E}[\mathbb{E}[Y_1 Y_2 | Y_1]] \text{ by Adam's law} \\ &= \mathbb{E}[Y_1 \mathbb{E}[Y_2 | Y_1]] \\ &= \mathbb{E} \left[Y_1 \left(\frac{1}{N-1} (N\mu - Y_1) \right) \right] \end{aligned}$$

The result above is because after we condition on Y_1 , we know $\mathbb{E}[Y_2 | Y_1]$ is the expected value over the remaining $Y_{2:n}$, which is just the sum of the remaining data, $N\mu - Y_1$, over $N - 1$. More algebra yields the following result:

$$\text{Var}(\bar{Y}) = \boxed{\frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)}$$

The factor above is called the finite population correction. Let's do an intuitive check. If $N = n$, then the $\text{Var}(\bar{Y}) = 0$. This makes sense, since this is the case of a census, meaning we can exactly know the population-level estimand. If $N \gg n$, then $\text{Var}(\bar{Y}) \sim \sigma^2/n$, which is the same variance from sampling with replacement. This makes sense because when a population is much larger than our sample, sampling with and without replacement is essentially equivalent.

The idea of finite population correction leads to an intuitive result. The absolute sample size n drives most of the variance in estimating μ , rather than the relative sample size. One way to think about this is to imagine one small bowl of soup and one large bowl, both well-mixed. We can take a tablespoon from either to taste, and that gives us a similar degree of confidence about how each soup tastes. We don't need a bigger spoon for the bigger bowl!

Now, let's move on to the third main type of sampling strategy, which is stratified random

sampling.

Definition: Stratified random sampling

In the stratified random sampling approach, we partition the population values $y_1, \dots, y_N$ into L subsets (strata) with sizes $N_1, \dots, N_L$. Then, we perform an SRS (with or without replacement) independently across strata, with corresponding sample sizes $n_1, \dots, n_L$.

But how do we choose the optimal $n_1, \dots, n_L$ for our strata? If we preset a bound n on the total sample size $n_1 + \dots + n_L = n$, it turns out that the optimal assignment is

$$n_\ell \propto N_\ell \sigma_\ell,$$

a strategy known as the Neyman allocation. This makes intuitive sense, as we would want to sample more from large strata and strata with high variance. However, σ_ℓ is unknown, which makes this difficult.

Let's work through an example of estimating the population mean with this new, stratified approach.

Example: Estimating population mean using stratified random sampling

We'd like to estimate the population μ , which can be written in terms of the stratum size N_ℓ and the stratum mean μ_ℓ :

$$\mu = \frac{1}{N} \sum_{\ell=1}^L N_\ell \mu_\ell$$

A natural estimator would be to replace the stratum mean with the sample mean for that stratum:

$$\hat{\mu}_{\text{strat}} = \frac{1}{N} \sum_{\ell=1}^L N_\ell \bar{Y}_\ell$$

Next, we can quickly derive the bias of $\hat{\mu}_{\text{strat}}$:

$$\mathbb{E}[\hat{\mu}_{\text{strat}}] = \frac{1}{N} \sum_{\ell=1}^L N_\ell \mathbb{E}[\bar{Y}_\ell] = \frac{1}{N} \sum_{\ell=1}^L N_\ell \mu_\ell = \mu \text{ by lin., symm} \Rightarrow \boxed{\hat{\mu}_{\text{strat}} \text{ is unbiased}}$$

as well as the variance:

$$\text{Var}(\hat{\mu}_{\text{strat}}) = \boxed{\frac{1}{N^2} \sum_{\ell=1}^L N_\ell^2 \text{Var}(\bar{Y}_\ell)},$$

where $\text{Var}(\bar{Y}_\ell)$ depends on if each stratum was sampled with/without replacement. We can simply plug in the results from the previous SRS examples.

Finally, we introduce a general estimator for μ that holds across many sampling designs.

Definition: Horvitz-Thompson estimator

Define the inclusion probability of a sampling design as $\pi_j = \mathbb{P}(j \in S)$, the probability of individual with ID j being included in the random sample S , which is a random set. Suppose $\pi_j > 0$ for all j (all individual can be sampled) and π_j is known. Then the Horvitz-Thompson estimator is an unbiased estimator of μ .

$$\hat{\mu}_{\text{HT}} = \frac{1}{N} \sum_{j=1}^N \frac{\mathbb{1}(j \in S)}{\pi_j} y_j$$

Proof:

$$\begin{aligned} \mathbb{E}[\hat{\mu}_{\text{HT}}] &= \frac{1}{N} \sum_{j=1}^N \frac{\mathbb{E}[\mathbb{1}(j \in S)]}{\pi_j} y_j \text{ by linearity} \\ &= \frac{1}{N} \sum_{j=1}^N \frac{\mathbb{P}(j \in S)}{\pi_j} y_j \\ &= \frac{1}{N} \sum_{j=1}^N y_j = \mu \end{aligned}$$

Note that the indicator means we only count the individuals in our sample! Also, the strategy of making the indicator cancel with the probability in the denominator is called inverse probability weighting.

10.3 Resampling

Performing hypothesis tests and studying estimators often involves parametric assumptions, asymptotics, and difficult calculations. Resampling allows us to carry out nonparametric methods that let the computer do all the work!

Say we want to compare the data collected for two groups. Within the frequentist framework, we saw that a common approach would be to perform a t-test, which would 1) assume Normal data, and 2) assume equal variance between groups. This procedure makes strong parametric assumptions! Instead, we will use a nonparametric approach that resamples the data we already have to compare the two distributions.

Definition: Two-sample permutation test

The permutation test tests the hypothesis that two sets of data have the same distribution. Suppose we have data $X_1, \dots, X_m \stackrel{\text{i.i.d.}}{\sim} F_X$ and $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} F_Y$, which represent

two independent samples. We define the hypotheses:

$$H_0 : F_X = F_Y \text{ vs. } H_1 : F_X \neq F_Y$$

Now, a permutation test is carried out as follows.

1. Define a test-statistic T (ex: difference in sample means)
2. Compute test-statistic t_0 from data
3. Compute T for each permutation of $X_1, \dots, X_m, Y_1, \dots, Y_n$ and call these $t_1, \dots, t_{(m+n)!}$
4. Calculate the p-value based on choosing a random permutation as

$$\mathbb{P}(T \geq t_0) = \frac{1}{(m+n)!} \sum_{j=1}^{(m+n)!} \mathbb{1}(t_j \geq t_0)$$

The idea is that under the null, both sets of data were generated under the same distribution $F_X = F_Y$. So, we can simply find the p-value of our data under the null of $F_X = F_Y$.

The two-sample permutation test is a clever, nonparametric way to use the data we already have without have collecting data in order to compare the distributions of two datasets.

Next, we introduce the bootstrap, a way to study the properties of an estimator $\hat{\theta}$ by resampling from the data we've already collected! Instead of finding an analytical form for the properties of $\hat{\theta}$ using asymptotics or expectation calculations, we let the computer do the work!

Definition: (Non-parametric) Bootstrap

Let our data be $\mathbf{y} = (y_1, \dots, y_n)$, assumed to be i.i.d. drawn from F . We'd like to study the properties of an estimator $\hat{\theta} = T(\mathbf{y})$. We can do this using non-parametric bootstrap sampling, defined below:

1. Generate the synthetic dataset $\mathbf{Y}_1^* = (Y_1^*, \dots, Y_n^*)$, which is a simple random sample from $\mathbf{y}$ with replacement.
2. Compute $\hat{\theta}_1^* = T(\mathbf{Y}_1^*)$, the value of our estimate from the synthetic data.
3. Repeat B times to obtain $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$.
4. Estimate properties of $\hat{\theta}$ using these simulated values:

$$\mathbb{E}[\hat{\theta}] \approx \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*, \text{ Var}(\hat{\theta}) \approx \frac{1}{B} \sum_{b=1}^B (\hat{\theta}_b^* - \bar{\theta}^*)^2$$

The core idea of the bootstrap is that since we don't know the distribution F , we are using the resampling using the ECDF $\hat{F}$ of our data $\mathbf{y}$ (equivalent to SRS with replacement). Since $\hat{F}(y) \xrightarrow{p} F(y)$ by LLN, we know that this is valid for large n . Here's a nice visual.

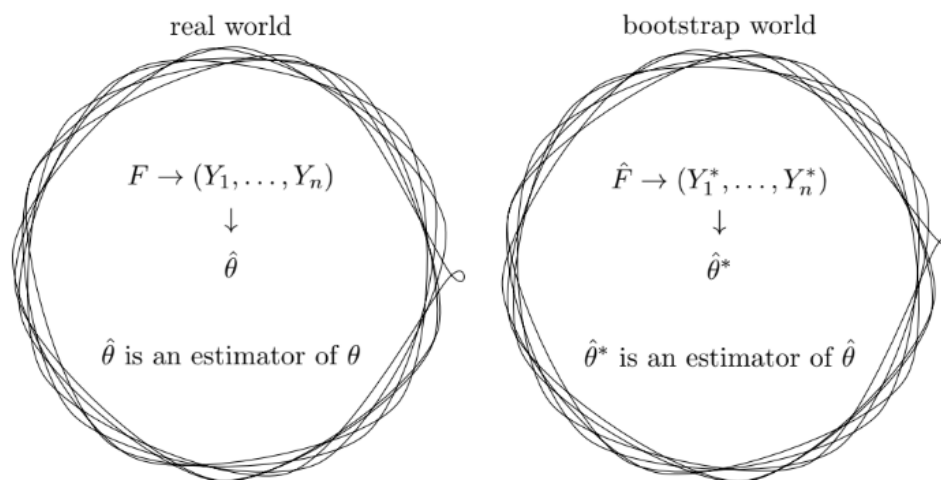


Figure 7: In the real world, data comes from a CDF F , and is then used to compute an estimator $\hat{\theta}$ for θ . In the bootstrap world, the data comes from the ECDF $\hat{F}$, and is then used to compute an estimator $\hat{\theta}^*$ for $\hat{\theta}$.

We can also apply this idea to generate bootstrap confidence intervals for our estimator $\hat{\theta}$.

Example: Bootstrap confidence intervals

Suppose we collect bootstrap estimates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ for $\hat{\theta}$. To generate a 95% confidence interval for θ , we can simply use the sample quantile of the $\hat{\theta}_j^*$.

$$[\hat{Q}_{\hat{\theta}^*}(0.025), \hat{Q}_{\hat{\theta}^*}(0.975)]$$

11 Causal inference

11.1 Causal framework

We will now discuss the framework for causal inference/experiments. Within the causal framework, we have units $j = 1, 2, \dots, n$, which are assigned a treatment.

Definition: Assignments, potential outcomes

For units $j = 1, \dots, n$, we define the treatment assignments as

$$\mathbf{W} = (W_1 \dots, W_n), \text{ where } W_j \in \{0, 1\}$$

Traditionally $W_j = 0$ means unit j is assigned a control, while $W_j = 1$ represents a treatment. Next, we define the potential outcome for each unit j :

$$Y_j(w_1, \dots, w_n)$$

Note that the potential outcome $Y_j(w_1, \dots, w_n)$ depends on the assignment of ALL units. This means that the treatment of unit 1 can impact the outcome of unit j . We can formalize the notion of treatment effect as well.

Definition: Treatment effect

We define the treatment effect of moving assignments from $w'_1, \dots, w'_n$ to $w_1, \dots, w_n$ as

$$\tau_j = Y_j(w_1, \dots, w_n) - Y_j(w'_1, \dots, w'_n)$$

This can be called the "causal effect of moving assignments". We can also define the average treatment effect (ATE) as

$$\bar{\tau} = \frac{1}{n} \sum_{j=1}^n \tau_j$$

We've seen the general case, where we assume the outcome of a single unit depends on the assignments of ALL units. Now, we simplify this framework by making the following assumption.

Definition: Non-interference

Under the non-interference assumption, we say that the j th potential outcome is only dependent on the j th assignment. Formally,

$$Y_j(w_{1:j-1}, w_j, w_{j+1:n}) = Y_j(w'_{1:j-1}, w_j, w'_{j+1:n})$$

for all $w_{1:n}$ and $w'_{1:j-1}, w'_{j+1:n}$. Under non-interference, we can also rewrite the treatment

effect in terms of only $Y_j(w_j)$.

$$\tau_j = Y_j(1) - Y_j(0)$$

Next, we formalize the notion of the assignment mechanism.

Definition: Assignment mechanism

Consider the following random variables:

1. Assignments : $\mathbf{W} = (W_1, \dots, W_n)$
2. Control potential outcomes : $\mathbf{Y}(0) = (Y_1(0), \dots, Y_n(0))$
3. Treatment potential outcomes : $\mathbf{Y}(1) = (Y_1(1), \dots, Y_n(1))$

Then, the assignment mechanism is specified by

$$\mathbb{P}(\mathbf{W} = \mathbf{w} | \mathbf{Y}(0), \mathbf{Y}(1))$$

The r.v.s $Y_1, \dots, Y_n$ can also be expressed in terms of their assignment W_j :

$$Y_j = Y_j(W_j) = W_j Y_j(1) + (1 - W_j) Y_j(0)$$

If we see observe $w_1 = 1$, meaning that $Y_j = Y_j(1)$, then $Y_j(0)$ is called the counterfactual, which is unobserved. This also means $\tau_j = Y_j(1) - Y_j(0)$ is unobserved.

11.2 Randomized controlled trial

We introduce the random controlled trial, a method of generating random assignments which are independent of the potential outcomes.

Definition: Randomized controlled trial

In an randomized controlled trial (RCT), we assign random assignments independent of everything.

$$\mathbb{P}(\mathbf{W} = \mathbf{w} | \mathbf{Y}(1), \mathbf{Y}(0)) = \mathbb{P}(\mathbf{W} = \mathbf{w})$$

In other words, $\mathbf{W}$ is independent of $\{\mathbf{Y}(0), \mathbf{Y}(1)\}$.

Suppose we construct an RCT such that we obtain the data

$$(w_1, y_1), \dots, (w_n, y_n)$$

Then, we can take two different routes for estimation.

1. Design-based estimand : $\bar{\tau}$

2. Population-based estimand : $\mathbb{E}[\tau_1]$ (assuming i.i.d. data)

With the design-based approach, we are focusing purely on the realized data we've collected. Here consider $\mathbf{W} | (\mathbf{Y}(1) = \mathbf{y}(1), \mathbf{Y}(0) = \mathbf{y}(0))$, similar to sampling theory. It means we are only able to draw conclusions about causal effects of the treatment on the n units in the study. It's a limited scope, but powerful, because we don't need to make any assumptions about the distribution of the data or how the individuals were sampled.

With the population-based approach, we are examining the overall treatment effect, meaning we are able to extrapolate outside of our data. Here, we consider $\mathbf{Y} | \mathbf{W} = \mathbf{w}$, conditioning on our assignments. However, we now have to impose assumptions about our data, such as a statistical model for our assignments and data, often assuming an i.i.d. parametric model.

11.3 Population-based models for experiments

We'll begin by exploring the population-based approach to experiments, where we focus solely on the data we've obtained!

In population-based modeling, we assume a statistical model for

$$\{W_1, Y_1(0), Y_1(1)\}, \dots, \{W_n, Y_n(0), Y_n(1)\}$$

which is independent across j , which also implies that our data come in independent pairs

$$(W_1, Y_1), \dots, (W_n, Y_n)$$

If we assume i.i.d. pairs and randomized assignments, then we can examine the population-based estimand.

$$\mathbb{E}[\tau_1] = \mathbb{E}[Y_1(1)] - \mathbb{E}[Y_1(0)]$$

Let's take this as an example.

Example: Average treatment effect for population-based model

Suppose we have i.i.d. pairs $(W_1, Y_1), \dots, (W_n, Y_n)$ and randomized assignments such that $\mathbf{W}$ and $\{\mathbf{Y}(0), \mathbf{Y}(1)\}$ are independent. Our estimand is

$$\mathbb{E}[\tau_1] = \mathbb{E}[Y_1(1)] - \mathbb{E}[Y_1(0)]$$

Let's begin by calculating the term $\mathbb{E}[Y_1(0)] = \theta_0$.

$$\begin{aligned} \theta_0 &= \mathbb{E}[Y_1(0)] = \mathbb{P}(Y_1(0) = 1) \\ &= \mathbb{P}(Y_1(0) = 1 | W_1 = 0) \text{ by RCT ind.} \\ &= \mathbb{P}(Y_1 = 1 | W_1 = 0) \end{aligned}$$

Using the same logic for $\mathbb{E}[Y_1(1)] = \theta_1$,

$$\theta_1 = \mathbb{E}[Y_1(1)] = \mathbb{P}(Y_1 = 1 | W_1 = 1)$$

Now, in order to estimate $\mathbb{E}[\tau_1]$ we can find the estimators $\hat{\theta}_0$ and $\hat{\theta}_1$, then subtract them! We've reduced this to a regression problem to solve for θ_0, θ_1 . Let's write down the likelihood function, where $\mathbf{W}$ is a predictor:

$$\begin{aligned}\log \mathcal{L}(\theta_0, \theta_1) &= \log \mathbb{P}(\mathbf{Y} = \mathbf{y} | \mathbf{W} = \mathbf{w}, \theta_0, \theta_1) \\ &= \sum_{j=1}^n \log \mathbb{P}(Y_j = y_j | \mathbf{W} = \mathbf{w}, \theta_0, \theta_1) \\ &= \sum_{j=1}^n \left\{ (1 - w_j) \log[\theta_0^{y_j} (1 - \theta_0)^{1-y_j}] + (w_j) \log[\theta_1^{y_j} (1 - \theta_1)^{1-y_j}] \right\}\end{aligned}$$

Then, we obtain the following results:

$$\hat{\theta}_1 = \frac{\sum_{j=1}^n Y_j w_j}{\sum_{j=1}^n w_j}, \quad \hat{\theta}_0 = \frac{\sum_{j=1}^n Y_j (1 - w_j)}{\sum_{j=1}^n (1 - w_j)}$$

Finally, we can estimate

$$\widehat{\mathbb{E}[\tau_1]} = \hat{\theta}_1 - \hat{\theta}_0$$

Note that conditioning on the treatments $\mathbf{W} = \mathbf{w}$:

$$\mathbb{E}[\widehat{\mathbb{E}[\tau_1]} | \mathbf{W} = \mathbf{w}] = \mathbb{E}[\tau_1]$$

11.4 Design-based approach for experiments

Next, we consider the design based approach, where we condition on the potential outcomes, considering the r.v. $\mathbf{W} | (\mathbf{Y}(0) = \mathbf{y}(0), \mathbf{Y}(1) = \mathbf{y}(1))$. This resembles sampling, where the population values are treated as "fixed". The advantage of this approach is that we make NO assumptions on the data-generating process (model-free approach). However, the tradeoff is that we cannot extrapolate, and can only speak of causal effects on the people in the study.

Let's consider a design-based estimand, the average treatment effect $\bar{\tau}$.

Theorem 11.5.1: Estimating the average treatment effect

We consider the average treatment effect as our estimand

$$\bar{\tau} = \frac{1}{n} \sum_{j=1}^n \{y_j(1) - y_j(0)\}$$

Then, the following MoM estimator is conditionally unbiased given $\{\mathbf{Y}(1) =$

$\mathbf{y}(1), \mathbf{Y}(0) = \mathbf{y}(0)\}$:

$$\hat{\tau}_{\text{MoM}}(\mathbf{W}) = \frac{1}{n} \sum_{j=1}^n \frac{W_j Y_j}{\mathbb{E}[W_j]} - \frac{(1 - W_j) Y_j}{\mathbb{E}[1 - W_j]}$$

Proof:

We can write $W_1 Y_1 = W_1 Y_1(1)$ and $(1 - W_1) Y_1 = (1 - W_1) Y_1(0)$. Next, we define

$$G_1 = \frac{W_1 Y_1(1)}{\mathbb{E}[W_1]}, \quad G_0 = \frac{(1 - W_1) Y_1(0)}{\mathbb{E}[1 - W_1]}$$

Taking the design-based approach, we can condition on the outcomes to obtain the conditional expectation of G_1, G_0 .

$$\begin{aligned} \mathbb{E}[G_1 | Y_1(0), Y_1(1)] &= \frac{y_1(1) \mathbb{E}[W_1 | Y_1(0), Y_1(1)]}{\mathbb{E}[W_1]} \\ &= \frac{y_1(1) \mathbb{E}[W_1]}{\mathbb{E}[W_1]} \text{ by randomization} \\ &= y_1(1) \end{aligned}$$

By the same logic, $\mathbb{E}[G_0 | Y_1(0), Y_1(1)] = y_1(0)$. Thus, we obtain the estimator above using the MoM principle.

This is a fascinating result, as it means we can obtain an unbiased estimate for the ATE without any model of the underlying process. We can also use a design-based approach to perform a hypothesis test.

Definition: Fisher null

We can also use a design-based approach to hypothesis testing. We define the Fisher null below:

$$H_0 : \tau_j = y_j(1) - y_j(0) = 0 \quad \forall j = 1, \dots, n$$

In other words, the Fisher null says that there is no treatment effect for all individuals. Under the Fisher null,

$$y_j = y_j(1) = y_j(0)$$

Definition: Randomization test

Using the information above for the Fisher null, we define the following test-statistic:

$$|\hat{\tau}_{\text{MoM}}(\mathbf{w})| = \left| \frac{1}{n} \sum_{j=1}^n \left\{ \frac{w_j y_j}{\mathbb{E}[W_1]} - \frac{(1-w_j)y_j}{\mathbb{E}[1-W_1]} \right\} \right|$$

Note that since we specify the assignment procedure for $\mathbf{W} = \mathbf{w}$, $\mathbb{E}[W_1]$ and $\mathbb{E}[1-W_1]$ are known. Then, reject the Fisher null if

$$|\hat{\tau}_{\text{MoM}}(\mathbf{w})| > Q_{\hat{\tau}_{\text{MoM}}(\mathbf{w})}(1-\alpha)$$

In practice, we obtain an estimate of the quantile function by simulating many random assignments $\mathbf{w}_1^*, \dots, \mathbf{w}_B^*$ and finding the value of the test-statistic for each draw.

$$\hat{\tau}_{\text{MoM}}(\mathbf{w}_1^*), \dots, \hat{\tau}_{\text{MoM}}(\mathbf{w}_B^*)$$

Then, we estimate the p-value by using the simulated distribution of the test-statistic:

$$\text{p-value} \approx \frac{1}{B} \sum_{b=1}^B \mathbb{1}(\hat{\tau}_{\text{MoM}}(\mathbf{w}_b^*) \geq \hat{\tau}_{\text{MoM}}(\mathbf{w}))$$

11.5 Observational studies

In an RCT, we performed a randomization by drawing $\mathbf{W} = \mathbf{w}$, then making causal conclusions from the data $(w_1, y_1), \dots, (w_n, y_n)$. In observational studies, we simply use existing data. We have no control over the assignments, and we take the current values and assignments for what they are. More formally our data takes the form

$$(W_1, Y_1, X_1), \dots, (W_n, Y_1, X_n),$$

where the X_j are called confounders or predictors. Now, the assignments also depend on the X_j .

$$\mathbb{P}(\mathbf{W} | \mathbf{Y}(1), \mathbf{Y}(0), \mathbf{X})$$

One assumption that simplifies our analysis is described below:

Definition: Unconfoundedness

The assignments are unconfounded if conditional on the predictors $\mathbf{X}$, the assignments are randomized (independent of the outcomes).

$$[\mathbf{W} \perp\!\!\!\perp \{\mathbf{Y}(1), \mathbf{Y}(0)\} | \mathbf{X}]$$

In other words,

$$\mathbb{P}(\mathbf{W} | \mathbf{Y}(1), \mathbf{Y}(0), \mathbf{X}) = \mathbb{P}(\mathbf{W} | \mathbf{X})$$

In an RCT, we impose randomization, meaning we know how assignments were generated. In an observational study, we don't know how assignments were made, so we can estimate the expected assignments, often under an unconfoundedness assumption that we must justify.

Definition: Propensity score

Suppose we have observational data where data are conditionally independent given the predictors.

$$(W_1, Y_1) | X_1, \dots, (W_n, Y_n) | X_n \text{ cond. ind.}$$

Then, under an unconfoundedness assumption,

$$\begin{aligned} \mathbb{P}(\mathbf{W} = \mathbf{w} | \mathbf{Y}(1), \mathbf{Y}(0), \mathbf{X} = \mathbf{x}) &= \mathbb{P}(\mathbf{W} = \mathbf{w} | \mathbf{X} = \mathbf{x}) \text{ by unconf.} \\ &= \prod_{j=1}^n \mathbb{P}(W_j = w_j | X_j = x_j) \end{aligned}$$

We define the propensity score of a datapoint as

$$\lambda_j(x) = \mathbb{P}(W_j = 1 | X_j = x) = \mathbb{E}[W_j | X_j = x]$$